

Post-Quantum Security of Keyed Sum of Permutations and Its Siblings

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Abstract. The rapid advancement of quantum computing poses significant challenges to the security of existing cryptographic constructions. Several constructions that are provably secure in the classical setting, e.g., the 3-round Luby–Rackoff, Even–Mansour, Keyed Sum of Permutations, become vulnerable when the adversary is granted quantum oracle access (the Q2 model). In contrast, when the adversary is restricted to classical oracle queries while retaining the ability to perform quantum computations locally (the Q1 model), such attacks no longer apply. In this paper, we investigate the Q1 security of the Keyed Sum of Permutations construction and two closely related variants - one employing identical permutations and another using a single key. We prove that all three constructions achieve $n/3$ -bit security in the Q1 model. In addition, for the same-key variant, we exhibit a key-recovery attack with matching complexity, thereby establishing the tightness of our security bound. For the remaining two constructions, we derive key-recovery attacks with complexity $2^{2n/3}$.

Keywords: Keyed Sum of Permutations · Post-Quantum Security · Q1 Model · Reprogramming Lemma · Resampling Lemma

1 Introduction

With the rapid progress of quantum cryptanalysis, assessing the security of cryptographic constructions under quantum attack models has become a central problem in modern cryptography. In the realm of public-key cryptography, quantum algorithms such as Shor’s [13] demonstrate that integer factorization and discrete logarithms can be solved in polynomial time, thereby undermining the security foundations of most widely deployed asymmetric primitives. Consequently, schemes whose security relies on these hardness assumptions are no longer considered secure in the presence of quantum adversaries. Symmetric cryptography is also affected, albeit in a more nuanced manner. Grover’s algorithm [10] accelerates brute-force search, including exhaustive key search, yielding a quadratic improvement over classical methods and effectively reducing the effective key strength by a factor of two. However, the impact of quantum computation extends beyond generic search speedups. For certain symmetric constructions, specialized quantum algorithms exploiting structural properties, e.g., period-finding techniques, enable significantly stronger attacks.

Two principal adversarial models are commonly considered: Q1 and Q2. In the Q2 model, the adversary is given full quantum access to both public components and the keyed construction, allowing it to issue superposition queries through quantum oracles and leverage powerful algorithms such as Simon’s and Grover’s. This strong capability often

enables polynomial-time attacks against many classical symmetric schemes, rendering them insecure in this setting. However, the Q2 model assumes the existence of a quantum interface to secret-key implementations, an assumption that is generally unrealistic in practical deployments. In contrast, the Q1 model more accurately reflects real-world scenarios. In this model, the adversary may query public, key-less components in superposition but can interact with the keyed construction only through classical queries, since practical systems typically expose only classical interfaces. Although the adversary may possess a quantum computer, it cannot realize a quantum oracle for the unknown keyed instance. This asymmetric access results in a hybrid attack setting that combines quantum information obtained from public components with limited classical observations of the secret-key algorithm. Consequently, standard quantum algorithms cannot be applied directly, and security proofs must carefully address the intricate dependencies between the two types of queries. Establishing tight security bounds in the Q1 model is therefore essential both for evaluating the post-quantum security of deployed symmetric cryptosystems and for advancing foundational proof techniques in post-quantum cryptography.

Related Works. In recent years, the cryptographic community has made substantial progress in analyzing the post-quantum security of symmetric constructions in the Q1 adversarial model. Jaeger et al. [11] first proved a Q1 security lower bound for the FX construction, demonstrating that it achieves $(k + n)/3$ -bit security against non-adaptive attacks. Their result matches the previously known attacks of Bonnetain et al. [7], thus confirming the tightness of the bound. In [2] Alagic et al. established that the Even-Mansour construction provides $n/3$ -bit Q1 security under adaptive attacks, while the offline Simon’s algorithm attack results on the Even-Mansour construction by Bonnetain et al. [7] show this bound is tight. In [3] Alagic et al. further extended this line of work to the Tweakable Even-Mansour construction, showing that it also achieves $n/3$ -bit security in the adaptive Q1 setting. In [4], Bai et al. conducted a systematic study of Key-Alternating Ciphers, establishing Q1 security guarantees for both 2-round and general t -round variants. In [5] Basak et al. proved that 3/4-round Key-Alternating Feistels (KAF) achieve tight $n/3$ -bit PRP/SPRP security in the Q1 setting when the round functions are modeled as a random oracle (even if identical). Recently Tan et al. [16] showed that sum of Even-Mansour SoEM21 construction achieves Q1 security up to $n/3$ -bits, which matches the earlier attacks due to Li. et al. [12], thereby demonstrating the tightness of the bound.

Our Contributions. In this paper we consider the Keyed Sum of Permutations (KSoP) [14], a random permutation based PRF that achieves the same provable security as the Sum of Even Mansour in the classical setting while being strictly simpler. Let $P_1, P_2 \in \mathcal{P}(n)$ be two independent random permutations. The Keyed Sum of Permutations (KSoP) construction is defined as:

$$\text{KSoP}_{k_1, k_2}[P_1, P_2](x) := P_1(x \oplus k_1) \oplus P_2(x \oplus k_2).$$

The single-key variant of KSoP, dubbed as KSoP21, is defined by

$$\text{KSoP21}_k[P_1, P_2](x) := P_1(x \oplus k) \oplus P_2(x \oplus k).$$

The single-permutation variant of KSoP, dubbed as KSoP12, is defined as

$$\text{KSoP12}_{k_1, k_2}[P](x) := P(x \oplus k_1) \oplus P(x \oplus k_2).$$

The pictorial representations of all the three constructions KSoP, KSoP21, and KSoP12 are depicted in Fig. 1, Fig. 2, and Fig. 3, respectively.

As our main contribution of the paper, we analyzed the post-quantum security of all the above mentioned three constructions and showed that all the three constructions achieve $n/3$ -bits security. To be precise, we demonstrate that for any adversary $\mathcal{A}$ that making q

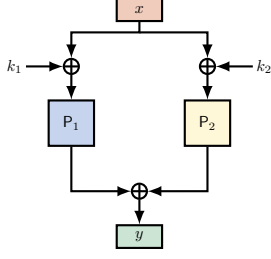


Figure 1: KSoP

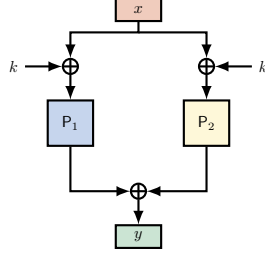


Figure 2: KSoP21

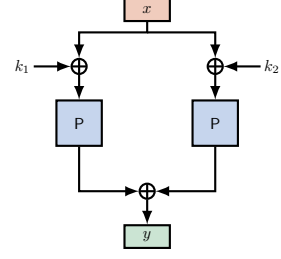


Figure 3: KSoP12

classical queries and total p quantum queries to its primitive oracle(s), the advantage of $\mathcal{A}$ is bounded by

$$O\left(\frac{q^2}{2^n} + \frac{p\sqrt{q} + q\sqrt{p}}{2^{n/2}}\right).$$

In addition, we have demonstrated a key-recovery attack for KSoP21 in $O(2^{n/3})$ data and time complexity, thereby obtaining tight post-quantum security for KSoP21. For KSoP and KSoP12, we show a key-recovery attack can be mounted in $O(2^{2n/3})$ data and time complexity.

TECHNIQUES. The techniques we employ are fairly generic and closely resemble those used in [16]. This suggests that classical security proofs for other similar (summation-type) cryptographic constructions, involving either a single permutation or on multiple independent permutations, can likely be adapted to the Q1 setting using similar arguments.

Our proofs build upon the techniques developed by Alagic et al. [2] in their post-quantum analysis of the Even–Mansour construction, suitably adapted to our setting. At a high level, the argument proceeds via a sequence of hybrid experiments in which both the classical construction queries and the quantum primitive queries are gradually replaced by their ideal counterparts. A central technical requirement is to ensure consistency between these two types of queries throughout the hybrid transitions.

Concretely, our analysis relies on two main tools: the Arbitrary Reprogramming (ARP) lemma and the Adaptive Resampling (ARS) lemma. The primary technical challenge arises from handling resampling at multiple points of a single underlying permutation. To address this issue, we extend the ARS lemma to accommodate resampling at multiple points, thereby enabling us to control the distinguishing advantage in this more general setting.

Organization. In Section 2 and Section 3, we present the necessary preliminaries and technical tools used throughout our analysis. We then state and prove the formal security results for KSoP, KSoP21 and KSoP12 in Section 4, Section 5, and Section 6, respectively (cf. Theorem 1, Theorem 2, and Theorem 3). Finally, in Section 7, we describe the key-recovery attacks on all three constructions.

2 Preliminaries

NOTATION. For any natural number n , $\{0, 1\}^n$ denotes the set of all bit strings of length n , and $\{0, 1\}^*$ denotes the set of all binary strings of arbitrary finite length. $x \xleftarrow{\$} X$ denotes that x is sampled uniformly at random from X . For any $n \in \mathbb{N}$, let $\mathcal{F}(n, n)$ denote the set of all functions $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, and let $\mathcal{P}(n) \subseteq \mathcal{F}(n, n)$ denote the set of all permutations on $\{0, 1\}^n$. Let $[n] = \{0, \dots, n\}$, for any natural number n . We denote an

empty set as $\emptyset$. For $X, Y \in \{0, 1\}^n$, we write $X \oplus Y$ to denote the bitwise xor of X and Y . For a function $F : \{0, 1\}^m \rightarrow \{0, 1\}^n$ and a set $B \subset \{0, 1\}^m \times \{0, 1\}^n$ such that each $x \in \{0, 1\}^m$ is the first element of at most one tuple in B , define

$$F^{(B)}(x) := \begin{cases} y, & \text{if } (x, y) \in B, \\ F(x), & \text{otherwise.} \end{cases}$$

For any $a, b, x \in \{0, 1\}^n$, we define the function $\text{swap}_{a,b}$ as follows:

$$\text{swap}_{a,b}(x) = \begin{cases} a, & x = b, \\ b, & x = a, \\ x, & \text{otherwise} \end{cases}$$

For any permutation $P \in \mathcal{P}(n)$, and any two uniformly chosen points $s_0, s_1 \in \{0, 1\}^n$, we denote the *reprogrammed* permutation of P w.r.t. (s_0, s_1) as $P^{(1)}$, which is defined as $P^{(1)} = P \circ \text{swap}_{s_0, s_1}$.

When an adversary has *quantum* access to a function $f : \{0, 1\}^m \rightarrow \{0, 1\}^n$, this means that the adversary accesses, as a black box, the $(m + n)$ qubit unitary operator $|x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle$.

Post-Quantum PRF Security. Let $\mathcal{F}(n, n)$ be the set of functions from $\{0, 1\}^n$ to $\{0, 1\}^n$. Let $\mathcal{A}^\mathcal{O}$ be an algorithm having access to an oracle $\mathcal{O}$ and making only *forward* queries $x \mapsto \mathcal{O}(x)$ to it, and outputting a bit. The *pseudorandom function* (PRF) advantage of $\mathcal{A}$ against a family of functions F_k , indexed by a secret key $k \in \{0, 1\}^\kappa$, is defined as:

$$\text{Adv}_F^{\text{PRF}}(\mathcal{A}) = \left| \Pr_{k \xleftarrow{\$} \{0, 1\}^\kappa} [\mathcal{A}^{F_k} \rightarrow 1] - \Pr_{F \xleftarrow{\$} \mathcal{F}(n, n)} [\mathcal{A}^F \rightarrow 1] \right|.$$

In addition, when $\mathcal{A}$ is a quantum algorithm (but access to $\mathcal{O}$ remains classical), we use the notation $\text{Adv}_F^{\text{pq-prf}}(\mathcal{A})$. We consider PQ-PRF definition when F_k is a *construction*, built over a *primitive* f that is modeled as a random function (or a random permutation). In that case, the primitive is chosen uniformly at random before running the game, and $\mathcal{A}$ accesses it via a quantum oracle $|x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle$. However, access to the construction remains classical.

3 Technical Tools

In this section, we present the main technical tools required for the Q1 security proofs developed in this paper. Most of these techniques are drawn from existing literature, although in several cases we adapt and refine them to suit our specific setting. For completeness and clarity, we restate the relevant results in a self-contained manner. These tools will play a central role in establishing the security bounds proved in the subsequent sections.

3.1 The Arbitrary Reprogramming Game

We recall the *arbitrary reprogramming lemma*, first introduced in [2]. It considers a distinguisher that first fixes a base oracle, along with a randomized procedure that outputs a set of points at which the oracle will be reprogrammed. The distinguisher is then given quantum access either to the original oracle or to the oracle modified at those specified points. The ARP lemma upper-bounds the distinguisher's advantage in distinguishing

between these two oracles. The adaptive reprogramming game for a distinguisher $\mathcal{D}$ is given as follows:

Phase 1: $\mathcal{D}$ outputs descriptions of a function $F_0 = F : \{0, 1\}^m \rightarrow \{0, 1\}^n$ and a randomized algorithm $\mathcal{B}$ whose output is a set $B \subset \{0, 1\}^m \times \{0, 1\}^n$ where each $x \in \{0, 1\}^m$ is the first element of at most one tuple in B . Let

$$B_1 = \{x \mid \exists y : (x, y) \in B\} \quad \text{and} \quad \epsilon = \max_{x \in \{0, 1\}^m} \left\{ \Pr_{B \leftarrow \mathcal{B}} [x \in B_1] \right\}.$$

Phase 2: $\mathcal{B}$ is run to obtain B . Let $F_1 = F^{(B)}$. A uniform bit b is chosen, and $\mathcal{D}$ is given quantum access to F_b .

Phase 3: $\mathcal{D}$ loses access to F_b , and receives the randomness r used to invoke $\mathcal{B}$ in Phase 2. Then $\mathcal{D}$ outputs a guess b' .

Lemma 1 (Reprogramming Lemma). *For any $\mathcal{D}$ making q queries in expectation when its oracle is F_0 , it holds that*

$$|\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1] - \Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0]| \leq 2q \cdot \sqrt{\epsilon}.$$

The proof of this lemma can be found in [2].

3.2 The Adaptive Resampling Game

In [2], Alagic et al. first introduced *the adaptive resampling lemma* (ARS) for random permutations, which is a generalization of the “reprogramming lemma” of Grilo et al. [9, Proposition 1]. It considers a distinguisher that is granted quantum access either to a uniformly random permutation P or to a modified permutation obtained by swapping the images of two distinct inputs under P . The ARS lemma provides an upper bound on the distinguisher’s advantage in distinguishing between these two. Formally, the ARS game for a distinguisher $\mathcal{D}$ is given as follows:

Phase 1: A permutation P is uniformly chosen from $\mathcal{P}(n)$, and $\mathcal{D}$ is given quantum access to $P^{(0)} = P$ and to its inverse.

Phase 2: $s_0, s_1 \in \{0, 1\}^n$ are uniformly chosen, and we let

$$P^{(1)} = P \circ \text{swap}_{s_0, s_1}.$$

A uniform bit $b \in \{0, 1\}$ is chosen, and $\mathcal{D}$ is given s_0, s_1 , and quantum access to $P^{(b)}$ and to its inverse. The $\mathcal{D}$ outputs a bit b' (try to guess b).

Lemma 2 (Resampling Lemma). *For any distinguisher $\mathcal{D}$, playing the above adaptive resampling game and making at most q queries (including both forward and inverse queries) to $P^{(0)}$ in the first phase, it holds that*

$$|\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1] - \Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0]| \leq 4 \cdot \sqrt{\frac{q}{2^n}}.$$

The proof of this lemma can be found in [2].

3.3 An Extended Adaptive Resampling Game

In our proofs, we must address the more general setting in which a single random permutation is reprogrammed via multiple *swap* operations. To capture this scenario, we extend

the ARS lemma to accommodate more than one swap. We formalize this generalized setting through the following extended resampling game defined for a distinguisher $\mathcal{D}$.

Phase 1: A permutation P is uniformly chosen from $\mathcal{P}_n$, and $\mathcal{D}$ is given quantum access to $P^{(0)} = P$.

Phase 2: $s_0, s_1, \dots, s_{2r-2}, s_{2r-1} \in \{0, 1\}^n$ are uniformly chosen, and we let

$$P^{(1)} = P \circ \prod_{i=1}^r \text{swap}_{s_{2i-2}, s_{2i-1}}.$$

A uniform bit $b \in \{0, 1\}$ is chosen, and $\mathcal{D}$ is given $s_0, s_1, \dots, s_{2r-2}, s_{2r-1}$, and quantum access to $P^{(b)}$. The $\mathcal{D}$ outputs a bit b' (try to guess b).

Lemma 3 (Extended Resampling Lemma). *For any $\mathcal{D}$ making at most q queries (including both forward and inverse queries) to $P^{(0)}$ in the first phase, it holds that*

$$|\Pr[\mathcal{D} \text{ outputs } 1 \mid b = 1] - \Pr[\mathcal{D} \text{ outputs } 1 \mid b = 0]| \leq 4\sqrt{\frac{rq}{2^n}},$$

where r is the number of swap operations used to reprogram the permutation.

Proof. The proof proceeds along the same lines as the proof of Lemma 2 presented by Alagic et al. [2], with the necessary modifications to accommodate our extended setting. For an arbitrary function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, define the state

$$|f\rangle_F = \bigotimes_{x \in \{0, 1\}^n} |f(x)\rangle_{F_x}.$$

Let $F = \{F_x\}_{x \in \{0, 1\}^n}$ be the collection of registers. The evaluation operator O acts as

$$O_{XYF} |x\rangle_X |y\rangle_Y |f\rangle_F = |x\rangle_X |y \oplus f(x)\rangle_Y |f\rangle_F.$$

For any permutation π , define

$$O_{XYF}^{\text{inv}} |\pi\rangle_F = \left(\sum_{x, y \in \{0, 1\}^n} |y\rangle \langle y|_Y \otimes X_X^x \otimes |y\rangle \langle y|_{F_x} \right) |\pi\rangle_F.$$

Here X is the Pauli- X operator and for $x = (x_1, \dots, x_n)$, $X^x := X^{x_1} \otimes \dots \otimes X^{x_n}$, and $X^x |\hat{x}\rangle = |\hat{x} \oplus x\rangle$. Then,

$$O_{XYF}^{\text{inv}} |x\rangle_X |y\rangle_Y |\pi\rangle_F = |x \oplus \pi^{-1}(y)\rangle_X |y\rangle_Y |\pi\rangle_F.$$

We may view a uniform permutation as a uniform superposition over all permutations in P_n . That is, define the state

$$|\phi_0\rangle_F = \frac{1}{\sqrt{(2^n)!}} \sum_{\pi \in P_n} |\pi\rangle_F.$$

The final state of any oracle algorithm $\mathcal{D}$ is identically distributed whether we:

- sample a uniform $\pi \in P_n$ and run $\mathcal{D}$ with access to π and π^{-1} , or
- run $\mathcal{D}$ with access to O and O^{inv} after initializing the F -registers to $|\phi_0\rangle_F$, and (if desired) measure the F -registers at the end to obtain π and the residual state of $\mathcal{D}$.

Now, we state the *Gentle Measurement Lemma* [1, Lemma 8], which is required for the proof.

Lemma 4 (Gentle Measurement Lemma). *Let $|\psi\rangle$ be a quantum state and let $\{P_i\}_{i=1}^q$ be a collection of projectors such that for all i , we have $\|\overline{P}_i |\psi\rangle\|_2^2 \leq \varepsilon_i$. Then,*

$$1 - |\langle\psi| (P_q \cdots P_1) |\psi\rangle|^2 \leq \sum_{i=1}^q \varepsilon_i.$$

where we denote the complementary projector of a projector P by $\overline{P} = \mathbb{I} - P$.

We split the distinguisher $\mathcal{D}$ into two stages $\mathcal{D} = (\mathcal{D}_0, \mathcal{D}_1)$ corresponding to the two phases of the experiment. We run the experiment using the superposition oracle $|\phi_0\rangle_F$ and measure the F -registers at the end.

On average over the choice of reprogramming positions $\{(s_{2i-2}, s_{2i-1})\}_{i=1}^r$, the adversary-oracle state after $\mathcal{D}_0$ is almost invariant under swapping the registers $F_{s_{2i-2}}$ and $F_{s_{2i-1}}$ unless $\mathcal{D}_0$ makes many oracle queries. Let $\tau = \{s_0, s_1, \dots, s_{2r-2}, s_{2r-1}\}$.

We define the projectors

$$(P_\tau)_X := \begin{cases} \mathbb{I} & \text{if } \exists u \neq v \text{ such that } s_u = s_v, \\ \mathbb{I} - (\sum_{i=0}^{2r-1} |s_i\rangle \langle s_i|) & \text{if } \forall u \neq v, s_u \neq s_v \end{cases}$$

$$(P_\tau^{\text{inv}})_{FY} := \begin{cases} \mathbb{I} & \text{if } \exists u \neq v \text{ such that } s_u = s_v, \\ \sum_{y \in \{0,1\}^n} |y\rangle \langle y|_Y \otimes (\mathbb{I} - |y\rangle \langle y|)_{F_{s_0} \dots F_{s_{2r-1}}}^{\otimes 2r} & \text{if } \forall u \neq v, s_u \neq s_v. \end{cases}$$

It is easy to verify that for any s_0, s_1 :

$$\left[\prod_{i=1}^r \text{Swap}_{F_{s_{2i-2}} F_{s_{2i-1}}}, O^{\text{XYF}}(P_\tau)_X \right] = 0, \quad (1)$$

$$\left[\prod_{i=1}^r \text{Swap}_{F_{s_{2i-2}} F_{s_{2i-1}}}, O_{\text{XYF}}^{\text{inv}}(P_\tau^{\text{inv}})_{FY} \right] = 0. \quad (2)$$

where $[\cdot, \cdot]$ denotes the commutator operation, and $\text{Swap}_{A,B}$ is the swap operator, i.e., $\text{Swap}_{A,B} |x\rangle_A |x'\rangle_B = |x'\rangle_A |x\rangle_B$ if A and B are distinct registers, and the identity if $A = B$.

Since $\overline{P}_\tau := \mathbb{I} - P_\tau \leq \sum_{i=0}^{2r-1} (|s_i\rangle \langle s_i|)$, it follows that for any normalized state $|\psi\rangle_{XE}$ (where E is an arbitrary register),

$$\mathbb{E}_{s_0, \dots, s_{2r-1}} \left[\|\overline{P}_\tau |\psi\rangle_{XE}\|_2^2 \right] \leq \mathbb{E}_{s_0, \dots, s_{2r-1}} \left[\langle\psi| \left(\sum_{i=0}^{2r-1} (|s_i\rangle \langle s_i|) \right) |\psi\rangle \right] = 2r \cdot 2^{-n}. \quad (3)$$

We show a similar statement for P_τ^{inv} . A valid adversary/oracle state $|\psi\rangle_{YXEF}$ (supported on the span of P_n) can be written as

$$|\psi\rangle_{YXEF} = \sum_{x, y \in \{0,1\}^n} c_{xy} |y\rangle_Y |y\rangle_{F_x} |\psi_{xy}\rangle_{XEF_{x^c}},$$

for normalized states $\{|\psi_{xy}\rangle\}_{x,y}$ satisfying $\sum_{x,y} |c_{xy}|^2 = 1$, and $\langle y|\psi_{xy}\rangle_{F_{x'}} = 0$, for all $x' \neq x$. If for any $i \neq j$, $s_i = s_j$, then

$$\|(\overline{P}_\tau^{\text{inv}})_{YF} |\psi\rangle_{YXEF}\|_2^2 = 0 \leq 2r \cdot 2^{-n}.$$

Hence, we have

$$\mathbb{E}_{s_0, \dots, s_{2r-1}} \left[\left\| (\bar{P}_\tau^{\text{inv}})_{YF} |\psi\rangle_{YXEF} \right\|_2^2 \right] \leq 2r \cdot 2^{-n}. \quad (4)$$

Without loss of generality, assume $\mathcal{D}_0$ starts in $|\psi_0\rangle = |\psi'_0\rangle |\phi_0\rangle$, and computes

$$|\psi\rangle = U_{\mathcal{D}_0} |\psi_0\rangle = U_q O_q U_{q-1} O_{q-1} \dots U_1 O_1 |\psi_0\rangle.$$

For every choice of $s_0, \dots, s_{2r-1}$, decompose $|\psi\rangle = |\psi_{\text{good}}(\tau)\rangle + |\psi_{\text{bad}}(\tau)\rangle$ and define

$$|\psi_{\text{good}}(\tau)\rangle = z U_q O_q P_\tau^q U_{q-1} O_{q-1} P_\tau^{q-1} \dots U_1 O_1 P_\tau^1 |\psi_0\rangle,$$

where

$$P_\tau^i = \begin{cases} P_\tau & \text{if } O_i = O, \\ P_\tau^{\text{inv}} & \text{if } O_i = O^{\text{inv}}, \end{cases}$$

and $z \in \mathbb{C}$ with $|z| = 1$ is chosen such that $\langle \psi | \psi_{\text{good}}(\tau) \rangle \in \mathbb{R}_{\geq 0}$. We can also view as

$$|\psi_{\text{good}}(\tau)\rangle = z U_{\mathcal{D}_0} \tilde{Q}_\tau^q \dots \tilde{Q}_\tau^1 |\psi_0\rangle,$$

with $\tilde{Q}_\tau^i = \tilde{U}_i^\dagger P_\tau^i \tilde{U}_i$, where $\tilde{U}_i = U_{i-1} O_{i-1} \dots U_1 O_1$. Now, define

$$\varepsilon_i(\tau) := \|\tilde{Q}_\tau^i |\psi_0\rangle\|_2^2 = \|\bar{P}_\tau^i \tilde{U}_i |\psi_0\rangle\|_2^2.$$

Then, by Lemma 4, we have

$$1 - |\langle \psi | \psi_{\text{good}}(\tau) \rangle|^2 \leq \sum_{i=1}^q \varepsilon_i(\tau). \quad (5)$$

Now we analyze the effect of reprogramming the (superposition) oracle after Phase 1. Consider $S := \prod_{i=1}^r \text{Swap}_{F_{s_{2i-2}}, F_{s_{2i-1}}}$. Here, reprogramming swaps the values of the permutation at s_{2i-2} and s_{2i-1} . Observe that $S|\phi_0\rangle = |\phi_0\rangle$. As the operator U_j s do not act on register F , from Eq. (1) and Eq. (2), we have

$$S|\psi_{\text{good}}(\tau)\rangle = |\psi_{\text{good}}(\tau)\rangle.$$

From the standard formula for the trace distance of pure states we get

$$\frac{1}{2} \|\psi\rangle\langle\psi| - S|\psi\rangle\langle\psi| S\|_1 = \sqrt{1 - |\langle\psi| S|\psi\rangle|^2}.$$

Now, we have

$$\begin{aligned} |\langle\psi| S|\psi\rangle| &= |\langle\psi_{\text{good}}(\tau)| S|\psi_{\text{good}}(\tau)\rangle + \langle\psi_{\text{good}}(\tau)| S|\psi_{\text{bad}}(\tau)\rangle \\ &\quad + \langle\psi_{\text{bad}}(\tau)| S|\psi_{\text{good}}(\tau)\rangle + \langle\psi_{\text{bad}}(\tau)| S|\psi_{\text{bad}}(\tau)\rangle| \\ &= |\langle\psi_{\text{good}}(\tau)| \psi_{\text{good}}(\tau)\rangle + \langle\psi_{\text{good}}(\tau)| \psi_{\text{bad}}(\tau)\rangle \\ &\quad + \langle\psi_{\text{bad}}(\tau)| \psi_{\text{good}}(\tau)\rangle + \langle\psi_{\text{bad}}(\tau)| S|\psi_{\text{bad}}(\tau)\rangle| \\ &= |\langle\psi|\psi\rangle + \langle\psi_{\text{bad}}(\tau)| (S - \mathbb{I})|\psi_{\text{bad}}(\tau)\rangle| \\ &\geq 1 - |\langle\psi_{\text{bad}}(\tau)| (S - \mathbb{I})|\psi_{\text{bad}}(\tau)\rangle| \\ &\geq 1 - 2\|\psi_{\text{bad}}(\tau)\|_2^2. \end{aligned}$$

Here, the above analysis follows from properties of Unitary operators and from the Cauchy-Schwarz and triangle inequalities. Thus, we have:

$$\frac{1}{2} \|\psi\rangle\langle\psi| - S|\psi\rangle\langle\psi| S\|_1 \leq 2\|\psi_{\text{bad}}(\tau)\|_2^2.$$

We know that $|\psi_{\text{bad}}(\tau)\rangle = |\psi\rangle - |\psi_{\text{good}}(\tau)\rangle$, therefore

$$\| |\psi_{\text{bad}}(\tau)\rangle \|_2^2 = 2 - 2 \cdot \text{Re}(\langle \psi | \psi_{\text{good}}(\tau) \rangle) = 2 - 2 \cdot |\langle \psi | \psi_{\text{good}}(\tau) \rangle| \leq 2 \sum_{i=1}^q \varepsilon_i(\tau).$$

Combining the last two results, we obtain

$$\frac{1}{2} \| |\psi\rangle \langle \psi| - S |\psi\rangle \langle \psi| S \|_1 \leq 2\sqrt{2} \sqrt{\sum_{i=1}^q \varepsilon_i(\tau)}. \quad (6)$$

Now, the rest of the proof is similar to that of the proof of [9, Theorem 6]. By the bound relating the distinguishing advantage and trace distance, we have

$$\begin{aligned} \Pr[\mathcal{D} \text{ outputs } b] - \frac{1}{2} &\leq \frac{1}{4} \cdot \mathbb{E}_{s_0, \dots, s_{2r-1}} [\| |\psi\rangle \langle \psi| - S |\psi\rangle \langle \psi| S \|_1] \\ &\leq \sqrt{2} \mathbb{E}_{s_0, \dots, s_{2r-1}} \left[\sqrt{\sum_{i=1}^q \varepsilon_i(\tau)} \right] \\ &\leq \sqrt{2} \sqrt{\mathbb{E}_{s_0, \dots, s_{2r-1}} \left[\sum_{i=1}^q \varepsilon_i(\tau) \right]} \\ &\leq 2 \sqrt{\frac{rq}{2^n}}, \end{aligned}$$

where above analysis follows from Eq. (3), Eq. (4), Eq. (6) and Jensen's inequality. $\square$

4 Post-Quantum Security of KSoP

In this section we prove the post-quantum security of the KSoP construction. Formally, the security result is as follows.

Theorem 1. *For any adversary $\mathcal{A}$ that makes q classical queries to KSoP, i.e., $\mathbb{E}[\mathbf{P}_1, \mathbf{P}_2]$ and p quantum queries to $\mathbf{P}_1, \mathbf{P}_2$ (and their inverses) in total, it holds that*

$$\text{Adv}_{\text{KSoP}}^{\text{pq-prf}}(\mathcal{A}) \leq \frac{q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.$$

The proof roughly goes as follows. We use a series of hybrids indexed by j ($\mathbf{H}_j$), where the first j classical queries are answered by random functions. The PRF advantage is then expressed depending on the probability to output 1, or to obtain a bad transcript, in all these games. For two consecutive games, to move from $\mathbf{H}_j$ to $\mathbf{H}_{j+1}$, we introduce an intermediate hybrid $\hat{\mathbf{H}}_j$. This is where we use first the reprogramming (Lemma 1), then the resampling (Lemma 2) bounds.

4.1 Proof of Theorem 1

Let $F \in \mathcal{F}(n, n)$ denote a uniform random function. Let $\mathcal{A}$ be a distinguisher and the goal of $\mathcal{A}$ is to distinguish between the *real world* ($\mathbb{E}[\mathbf{P}_1, \mathbf{P}_2], \mathbf{P}_1, \mathbf{P}_2$) and the *ideal world* ($F, \mathbf{P}_1, \mathbf{P}_2$). In the real world, the adversary $\mathcal{A}$ has *classical* access to the construction $\mathbb{E}[\mathbf{P}_1, \mathbf{P}_2]$, and *quantum* access to the permutations $\mathbf{P}_1, \mathbf{P}_2$ in both the forward and inverse directions. On the other hand, in the ideal world $\mathcal{A}$ has *classical* access to a random function F and *quantum* access to the permutations $\mathbf{P}_1, \mathbf{P}_2$ in both the forward and inverse directions.

Without loss of generality, we assume that $\mathcal{A}$ never makes any redundant (classical) construction query. Let the adversary $\mathcal{A}$ makes q many (classical) construction queries and p many (quantum) primitive queries in total. We divide the execution of $\mathcal{A}$ into $q + 1$ stages $0, \dots, q$ where the j -th stage corresponds to the time between j -th and $(j + 1)$ -th classical queries of $\mathcal{A}$. We allow $\mathcal{A}$ to adaptively distribute its p quantum queries between these stages arbitrarily. We let p_j denote the number of quantum queries made in the j -th stage. We denote the j -th classical query of $\mathcal{A}$ by (x_j, y_j) , that means $\mathcal{A}$ queries x_j to the construction and receives response y_j . Let $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$ be the ordered list describing the first j many classical queries made by $\mathcal{A}$.

We use “ $\prod$ ” to denote sequential composition of operations, i.e., $\prod_{i=1}^n f_i = f_1 \circ \dots \circ f_n$. Note that order matters, since in general composition of operators is not commutative. Now, we define the reprogrammed permutations corresponding to P_1 and P_2 respectively. For any $i \in [j]$, we sequentially choose random values $u_i \in \{0, 1\}^n \setminus (\mathcal{U} \cup \mathcal{Y} \cup P_1 \cup P_2)$, where $\mathcal{U} = \{u_1, \dots, u_{i-1}\}$, $\mathcal{Y} = \{y_i \oplus u_1 \oplus y_1, \dots, y_i \oplus u_{i-1} \oplus y_{i-1}\}$, $P_1 = \{P_1(x_1 \oplus k_1), \dots, P_1(x_i \oplus k_1)\}$, and $P_2 = \{P_2(x_1 \oplus k_2) \oplus y_i, \dots, P_2(x_i \oplus k_2) \oplus y_i\}$. Now, the corresponding reprogramming operators are defined as

$$\vec{S}_{T_j, P_1, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_1(x_i \oplus k_1), u_i}, \quad \vec{S}_{T_j, P_2, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_2(x_i \oplus k_2), (y_i \oplus u_i)}.$$

We define the reprogrammed permutations $P_{1, T_j, k}$ and $P_{2, T_j, k}$ by

$$P_{1, T_j, k} = \vec{S}_{T_j, P_1, k} \circ P_1, \quad P_{2, T_j, k} = \vec{S}_{T_j, P_2, k} \circ P_2.$$

Now, let $\mathbf{H}_0, \dots, \mathbf{H}_q$ denote a sequence of $q + 1$ hybrids, where $\mathbf{H}_j$ is defined as follows:

1. Sample $k = (k_1, k_2) \xleftarrow{\$} \{0, 1\}^{2n}$.
2. Answer the first j classical queries of $\mathcal{A}$ using F and the quantum queries of $\mathcal{A}$ in the first j stages (which ends when the $(j + 1)$ -th classical query is made) using P_1, P_2 .
3. Answer all subsequent classical queries from $\mathcal{A}$ using $E_k[P_1, P_2]$ and all subsequent quantum queries from $\mathcal{A}$ using $P_{1, T_j, k}$ and $P_{2, T_j, k}$.

Since, $T_0 = \emptyset$, we deduce that $\mathbf{H}_0$ corresponds to $\mathcal{A}$'s interaction with the real world. Similarly, one can easily notice that $\mathbf{H}_q$ corresponds to $\mathcal{A}$'s interaction with the ideal world. Thus, we have

$$\text{Adv}_{\text{KSoP}}^{\text{pq-prf}}(\mathcal{A}) = |\Pr[\mathcal{A}(\mathbf{H}_0) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_q) \rightarrow 1]|. \quad (7)$$

Now for any $j \in \{0, \dots, q\}$, $\mathbf{H}_j$ can be viewed as a string of oracle calls made by $\mathcal{A}$

$$\underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1, T_j, k}, P_{2, T_j, k})^{q-j}}_{q-j \text{ classical queries}}.$$

where each occurrence of F (res. $E_k[P_1, P_2]$) indicates a single classical query and each occurrence of P_1, P_2 (res. $P_{1, T_j, k}, P_{2, T_j, k}$) indicates the sequence of quantum primitive queries. Now, in this representation we want to compare

$$\begin{aligned} \mathbf{H}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1, P_2], P_{1, T_j, k}, P_{2, T_j, k}, (E_k[P_1, P_2], P_{1, T_j, k}, P_{2, T_j, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{H}_{j+1} &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, F, P_1, P_2, \underbrace{(E_k[P_1, P_2], P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

To aid this comparison we introduce an intermediate hybrid $\hat{\mathbf{H}}_j$ which is defined as follows:

1. Sample $k = (k_1, k_2) \xleftarrow{\$} \{0, 1\}^{2n}$.
2. Answer the first $j + 1$ classical queries of $\mathcal{A}$ using F and the quantum queries of $\mathcal{A}$ in the first j stages (which ends when the $(j + 1)$ -th classical query is made) using P_1, P_2 .
3. Answer all subsequent classical queries from $\mathcal{A}$ using $E_k[P_1, P_2]$ and all subsequent quantum queries from $\mathcal{A}$ using $P_{1,T_{j+1},k}$ and $P_{2,T_{j+1},k}$.

Now, we compare the following hybrids in pairs from top to bottom:

$$\begin{aligned}
\mathbf{H}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1, P_2], P_{1,T_j,k}, P_{2,T_j,k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_j,k}, P_{2,T_j,k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\
\hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{F, P_{1,T_{j+1},k}, P_{2,T_{j+1},k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\
\mathbf{H}_{j+1} &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{F, P_1, P_2}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}.
\end{aligned}$$

In the following, we state two important lemmas - Lemma 5 and Lemma 6 that establish the bounds on the distinguishability between $\hat{\mathbf{H}}_j$ and $\mathbf{H}_{j+1}$, between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, respectively. The proofs of the lemmas are given in Section 4.2 and Section 4.3.

Lemma 5. For $j = 0, \dots, q - 1$, we have

$$| \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1] | \leq 2p_{j+1} \sqrt{\frac{2(j+1)}{2^n}},$$

where p_{j+1} is the expected number of quantum queries $\mathcal{A}$ makes in the $(j + 1)$ -th stage of the game.

Lemma 6. For $j = 0, \dots, q - 1$, we have

$$| \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \leq \frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}},$$

where q and p are the expected number of classical and quantum queries made by $\mathcal{A}$.

Given the above two lemmas and Eq. (7), we have

$$\begin{aligned}
\text{Adv}_{\text{KSOP}}^{\text{pq-prf}}(\mathcal{A}) &= | \Pr[\mathcal{A}(\mathbf{H}_0) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_q) \rightarrow 1] | \\
&\leq \sum_{j=0}^{q-1} \left(| \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \right. \\
&\quad \left. + | \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1] | \right) \\
&\leq \sum_{j=0}^{q-1} \left(\frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}} + 2p_{j+1} \sqrt{\frac{2(j+1)}{2^n}} \right) \\
&\leq \frac{q^2}{2^n} + 12q\sqrt{\frac{p}{2^n}} + 2p\sqrt{\frac{2q}{2^n}} \\
&\leq \frac{q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.
\end{aligned}$$

4.2 Proof of Lemma 5

Recall we can have the following string of oracle call representations for the two hybrids.

$$\begin{aligned} \hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \quad \mathbf{F}, \quad \mathbf{P}_{1,T_{j+1},k}, \quad \mathbf{P}_{2,T_{j+1},k}, \quad \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{H}_{j+1} &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \quad \mathbf{F}, \quad \mathbf{P}_1, \quad \mathbf{P}_2, \quad \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

It is clear that the two hybrids are identical, except for the $(j+1)$ -th stages, where $\hat{\mathbf{H}}_j$ uses reprogrammed versions of the permutations P_1 and P_2 , and $\mathbf{H}_{j+1}$ uses the original permutations P_1 and P_2 . We will show that the distinguishing advantage of these two hybrids can be captured by the *reprogramming* bounds of the permutations.

More formally, given the adversary $\mathcal{A}$ we construct an ARP distinguisher $\mathcal{D}$ as follows (note that Phases 1, 2, 3 correspond to Phases 1, 2, 3 in the ARP game respectively).

Phase 1. For all the queries up to and including the $(j+1)$ -th classical query:

1. $\mathcal{D}$ samples uniform random permutations $P_1, P_2 \leftarrow \mathcal{P}(n)$ and uniform random function $F \leftarrow \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_{j+1} = \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 . More formally, $\mathcal{D}$ defines $F(\alpha, t, x) := P_\alpha^t(x)$, for $t \in \{1, -1\}$ and $\alpha \in \{1, 2\}$.
4. $\mathcal{D}$ defines the randomized algorithm $\mathcal{B}$, which takes a key k , computes a set B of input/output pairs to be reprogrammed so that $F^{(B)}(\alpha, t, x) = P_{\alpha, T_{j+1}, k}^t(x)$, for all α, t, x .

Phase 2. For all the quantum queries in the $(j+1)$ -th stage:

1. $\mathcal{B}$ is run to generate the set B and $\mathcal{D}$ is given quantum access to F_b , where $F_0 = F(= P_\alpha^t)$ and $F_1 = F^{(B)}(= P_{\alpha, T_{j+1}, k}^t)$.
2. All quantum queries from $\mathcal{A}$ are answered using F_b .

Phase 3. For all the queries beyond and including $(j+2)$ -th classical query:

1. $\mathcal{D}$ is given the randomness (specifically the key k here) used by $\mathcal{B}$ to generate B .
2. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1, P_2]$.
3. All quantum queries from $\mathcal{A}$ are answered using F_1 .
4. Finally, $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARP distinguisher. If $b = 0$ (i.e., $\mathcal{D}$'s oracle in **Phase 2** is $F_0 = F$), then the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_{j+1}$. On the other hand, if $b = 1$ (i.e., $\mathcal{D}$'s oracle in **Phase 2** is $F_1 = F^{(B)}$), then the output of $\mathcal{A}$ is identically distributed to its output in $\hat{\mathbf{H}}_j$. It follows that $|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]|$ is equal to the distinguishing advantage of $\mathcal{D}$ in the reprogramming game. To bound this quantity using [Lemma 1](#), we bound the reprogramming probability ϵ and the expected number of queries made by $\mathcal{D}$ in **Phase 2** (when $F_0 = F$).

The reprogramming probability ϵ can be bounded using the definition of $P_{\alpha, T_{j+1}, k}$ and the fact that $F^{(B)}(\alpha, t, x) = P_{\alpha, T_{j+1}, k}^t(x)$. Fixing P_1, P_2 and T_{j+1} , the probability that any given (α, t, x) is reprogrammed is at most the probability (over k) that it is in the set:

$$\{(1, 1, (x_i \oplus k_1)), (1, -1, P_1^{-1}(u_i)), (2, 1, (x_i \oplus k_2)), (2, -1, P_2^{-1}(y_i \oplus u_i))\}_{i=1}^{j+1}$$

Here, for any (α, t, x) , either it is a forward query or a backward query to F . In both cases, probability that (α, t, x) is reprogrammed is upper bounded by $2(j+1)/2^n$. Hence, we get $\epsilon \leq 2(j+1)/2^n$.

The expected number of queries made by $\mathcal{D}$ in **Phase 2** when $F_0 = F$ is equal to the expected number of queries made by $\mathcal{A}$ in its $(j+1)$ -th stage, which is p_{j+1} . Thus, by **Lemma 1**, we have

$$|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \leq 2p_{j+1} \sqrt{\frac{2(j+1)}{2^n}}.$$

4.3 Proof of Lemma 6

For bounding the distinguishing advantage between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, we will introduce two intermediate hybrids called $\mathbf{\Gamma}_j$, $\mathbf{\Sigma}_j$. Now, we bound the distinguishing advantages between the following hybrids in pairs from top to bottom.

$$\begin{aligned} \mathbf{H}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1, P_2]}_{q-j-1 \text{ classical queries}}, \underbrace{P_{1,T_j,k}, P_{2,T_j,k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_j,k}, P_{2,T_j,k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{\Gamma}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1^{(1)}, P_2]}_{q-j-1 \text{ classical queries}}, \underbrace{P_{1,T_j,k}^{(1)}, P_{2,T_j,k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1^{(1)}, P_2], P_{1,T_j,k}^{(1)}, P_{2,T_j,k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{\Sigma}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1^{(1)}, P_2^{(1)}]}_{q-j-1 \text{ classical queries}}, \underbrace{P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1^{(1)}, P_2^{(1)}], P_{1,T_j,k}^{(1)}, P_{2,T_j,k}^{(1)})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{F}_{q-j-1 \text{ classical queries}}, \underbrace{P_{1,T_{j+1},k}, P_{2,T_{j+1},k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

Let, $\mathcal{A}$ be an adversary of $\mathbf{H}_j$ and $\mathbf{\Gamma}_j$. From $\mathcal{A}$, we can construct an ARS distinguisher $\mathcal{D}$ as follows.

Phase 1. For all the queries up to (immediately before) the $(j+1)$ -th classical query:

1. $\mathcal{D}$ is given quantum oracle access to permutation P_1 and $\mathcal{D}$ samples an independent random permutation $P_2 \in \mathcal{P}(n)$ and uniform random function $F \in \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 .

Phase 2. For all the queries beyond and including $(j+1)$ -th classical query:

1. $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ and $\mathcal{D}$ has now quantum oracle access to $P_1^{(b)}$, where $P_1^{(0)} = P_1$ and $P_1^{(1)} = \text{swap}_{P_1(s_0), P_1(s_1)} \circ P_1$.
2. Let $\mathcal{D}$ receives the $(j+1)$ -th classical query from $\mathcal{A}$ which is x_{j+1} .
3. $\mathcal{D}$ sets $k_1 = x_{j+1} \oplus s_0$ and $k_2 = x_{j+1} \oplus s_1$, and $k = (k_1, k_2)$.
4. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1^{(b)}, P_2]$.
5. All quantum queries from $\mathcal{A}$ are answered using $P_{1,T_j,k}^{(b)}$ and $P_{2,T_j,k}$ correspondingly.
6. Finally $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARS distinguisher. If $b = 0$, then $P_1^{(0)} = P_1$, thus the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_j$. On the other hand, if $b = 1$, then the output of $\mathcal{A}$ is identically distributed to its output in $\mathbf{\Gamma}_j$. Hence, the distinguishing advantage between the hybrids $\mathbf{H}_j$ and $\mathbf{\Gamma}_j$ is equal to the distinguishing advantage of $\mathcal{D}$ in the resampling game. Using Lemma 2 we have

$$|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{\Gamma}_j) \rightarrow 1]| \leq 4\sqrt{\frac{p}{2^n}}. \quad (8)$$

Now, we bound the distinguishing advantage between the hybrids $\mathbf{\Gamma}_j$ and $\mathbf{\Sigma}_j$. Let $\mathcal{A}$ be an adversary between these two hybrids, we can construct an ARS distinguisher $\mathcal{D}$ as follows. **Phase 1.** For all the queries up to (immediately before) the $(j+1)$ -th classical query:

1. $\mathcal{D}$ is given quantum oracle access to permutation P_2 and $\mathcal{D}$ samples an independent random permutation $P_1 \in \mathcal{P}(n)$ and a random function $F \in \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 .

Phase 2. For all the queries beyond and including $(j+1)$ -th classical query:

1. $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ and $\mathcal{D}$ has now quantum oracle access to $P_2^{(b)}$, where $P_2^{(0)} = P_2$ and $P_2^{(1)} = \text{swap}_{P_2(s_0), P_2(s_1)} \circ P_2$.
2. Let $\mathcal{D}$ receives the $(j+1)$ -th classical query from $\mathcal{A}$ which is x_{j+1} .
3. $\mathcal{D}$ sets $k_1 = x_{j+1} \oplus s_0$ and $k_2 = x_{j+1} \oplus s_1$, and $k = (k_1, k_2)$. $\mathcal{D}$ reprogram the permutation P_1 to get $P_1^{(1)}$.
4. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1^{(1)}, P_2^{(b)}]$.
5. All quantum queries from $\mathcal{A}$ are answered using $P_{1, T_j, k}^{(1)}$ and $P_{2, T_j, k}^{(b)}$ correspondingly.
6. Finally $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

Clearly, $\mathcal{D}$ is a valid ARS distinguisher. If $b = 0$, then $P_2^{(0)} = P_2$, thus the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{\Gamma}_j$. On the other hand, if $b = 1$, then the output of $\mathcal{A}$ is identically distributed to its output in $\mathbf{\Sigma}_j$. Hence, the distinguishing advantage between the hybrids $\mathbf{\Gamma}_j$ and $\mathbf{\Sigma}_j$ is equal to the distinguishing advantage of $\mathcal{D}$ in the resampling game. Using Lemma 2 we have

$$|\Pr[\mathcal{A}(\mathbf{\Gamma}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{\Sigma}_j) \rightarrow 1]| \leq 4\sqrt{\frac{p}{2^n}}. \quad (9)$$

Now, consider the following two events:

1. **bad₁**: $(s_1 \oplus k_1) \in \{x_1, \dots, x_j\}$ or $(s_0 \oplus k_2) \in \{x_1, \dots, x_j\}$.
2. **bad₂**: In phase 2, $\mathcal{A}$ queries its classical oracle on $s_1 \oplus k_1$ or on $s_0 \oplus k_2$ i.e $s_1 \oplus k_1 \in \{x_{j+2}, \dots, x_q\}$ or $s_0 \oplus k_2 \in \{x_{j+2}, \dots, x_q\}$.

Since, $s_1 \oplus k_1 = s_0 \oplus k_2 = s_1 \oplus s_0 \oplus x_{j+1}$ is uniform, $\Pr[\text{bad}_1] \leq j/2^n$. Now for the event **bad₂** consider the distinguisher $\mathcal{D}'$ that behaves identically to $\mathcal{D}$ in Phase 1 and Phase 2, but then when $\mathcal{A}$ terminates outputs 1 iff event **bad₂** occurred. When $b = 0$, the probability that **bad₂** occurs is at most $(q-j)/2^n$. Since, $\mathcal{D}'$ is a distinguisher for the resampling game, so, we get $\Pr[\text{bad}_2] \leq (q-j)/2^n + 4\sqrt{p/2^n}$.

Algorithm 1 Hybrid experiment Σ_j . Including the boxed statements results in Σ'_j .

```

1:  $P_1, P_2 \xleftarrow{\$} \mathcal{P}(n), F \xleftarrow{\$} \mathcal{F}(n, n)$ 
2: Run  $\mathcal{A}$  with quantum access to  $P_1, P_2$  and classical access to  $F$ , until  $\mathcal{A}$  makes its
    $(j+1)$ -th classical query  $x_{j+1}$ 
3:  $T_j := \{(x_1, y_1), \dots, (x_j, y_j)\}$ 
4:  $s_0, s_1 \xleftarrow{\$} \{0, 1\}^n, P_1^{(1)} := P_1 \circ \text{swap}_{s_0, s_1}, P_2^{(1)} := P_2 \circ \text{swap}_{s_0, s_1}$ 
5:  $k_1 := s_0 \oplus x_{j+1}, k_2 := s_1 \oplus x_{j+1}, k := (k_1, k_2)$ 
6:  $y_{j+1} := E_k[P_1^{(1)}, P_2^{(1)}](x_{j+1})$ 
7:  $Q_1 := P_{1, T_j, k}^{(1)}, Q_2 := P_{2, T_j, k}^{(1)}$ 
8: Give  $y_{j+1}$  to  $\mathcal{A}$  as the answer to its  $(j+1)$ -th classical query
9:  $T_{j+1} := \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$ 
10:  $u_{j+1} := P_1(s_1)$ 
11: if  $(s_1 \oplus k_1) \in \{x_1, \dots, x_j\} \vee (s_0 \oplus k_2) \in \{x_1, \dots, x_j\}$  then
12:    $\text{bad}_1 := \text{true}$ 
13:    $Q_1 := P_{1, T_{j+1}, k}$  and  $Q_2 := P_{2, T_{j+1}, k}$ 
14: Continue running  $\mathcal{A}$  with quantum access to  $Q_1, Q_2$ 
   (The remaining classical queries are answered as follows:)
15:  $y := E_k[P_1^{(1)}, P_2^{(1)}](x)$ 
16: if  $x = (s_1 \oplus k_1) \vee x = (s_0 \oplus k_2)$  then
17:    $\text{bad}_2 := \text{true}$ 
18:    $y := E_k[P_1, P_2]$ 
19: return  $y$ .
```

In Algorithm 1, we described Σ_j and Σ'_j . Observe that, Σ_j and Σ'_j are identical until either bad_1 or bad_2 occur, and so we have

$$|\Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma'_j) \rightarrow 1]| \leq \Pr[\text{bad}_1 \vee \text{bad}_2] \leq \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}}. \quad (10)$$

Now, we are arguing that Σ'_j is identical to $\hat{\mathbf{H}}_j$:

1. Up to (just before) the $(j+1)$ -th classical query Σ'_j is exactly identical to $\hat{\mathbf{H}}_j$.
2. In Σ'_j , the response of the $(j+1)$ -th classical query, y_{j+1} is defined as

$$\begin{aligned} y_{j+1} &= E_k[P_1^{(1)}, P_2^{(1)}](x_{j+1}) = P_1^{(1)}(x_{j+1} \oplus k_1) \oplus P_2^{(1)}(x_{j+1} \oplus k_2) \\ &= P_1^{(1)}(s_0) \oplus P_2^{(1)}(s_1) = P_1(s_1) \oplus P_2(s_0). \end{aligned} \quad (11)$$

So, the value y_{j+1} is uniformly distributed.

3. In Σ'_j , the oracles Q_1 and Q_2 are always equal to $P_{1, T_{j+1}, k}$ and $P_{2, T_{j+1}, k}$ respectively. When bad_1 occurs this is immediate (since then Q_1 and Q_2 are set to $P_{1, T_{j+1}, k}$ and $P_{2, T_{j+1}, k}$ resp. in line 13). But if bad_1 does not occur then for $i = 1, \dots, j$ we have $x_i \oplus k_1 \neq s_0$ and $x_i \oplus k_1 \neq s_1$ (where the former is because $x_{j+1} \oplus k_1 = s_0$ but $x_i \neq x_{j+1}$ by assumption, and the latter is by definition of bad_1). Similarly, for $i = 1, \dots, j$ we have $x_i \oplus k_2 \neq s_0$ and $x_i \oplus k_2 \neq s_1$. So, $P_1^{(1)}(x_i \oplus k_1) = P_1(x_i \oplus k_1)$ and $P_2^{(1)}(x_i \oplus k_2) = P_2(x_i \oplus k_2)$, for $i = 1, \dots, j$. Thus

$$\vec{S}_{T_j, P_1^{(1)}, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_1^{(1)}(x_i \oplus k_1), u_i} = \prod_{i=1}^j \text{swap}_{P_1(x_i \oplus k_1), u_i} = \vec{S}_{T_j, P_1, k}$$

and

$$\vec{S}_{T_j, P_2^{(1)}, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_2^{(1)}(x_i \oplus k_2), (y_i \oplus u_i)} = \prod_{i=1}^j \text{swap}_{P_2(x_i \oplus k_2), (y_i \oplus u_i)} = \vec{S}_{T_j, P_2, k}$$

Therefore,

$$\begin{aligned} Q_1 &= P_{1, T_j, k}^{(1)} = \vec{S}_{T_j, P_1^{(1)}, k} \circ P_1^{(1)} = \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(s_0), P_1(s_1)} \circ P_1 \\ &= \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(x_{j+1} \oplus k_1), P_1(s_1)} \circ P_1 \\ &= \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(x_{j+1} \oplus k_1), u_{j+1}} \circ P_1 \\ &= \vec{S}_{T_{j+1}, P_1, k} \circ P_1 \\ &= P_{1, T_{j+1}, k} \end{aligned}$$

and

$$\begin{aligned} Q_2 &= P_{2, T_j, k}^{(1)} = \vec{S}_{T_j, P_2^{(1)}, k} \circ P_2^{(1)} = \vec{S}_{T_j, P_2, k} \circ \text{swap}_{P_2(s_0), P_2(s_1)} \circ P_2 \\ &= \vec{S}_{T_j, P_2, k} \circ \text{swap}_{P_2(s_0), P_2(x_{j+1} \oplus k_2)} \circ P_2 \\ &= \vec{S}_{T_j, P_2, k} \circ \text{swap}_{(u_{j+1} \oplus y_{j+1}), P_2(x_{j+1} \oplus k_2)} \circ P_2 \\ &= \vec{S}_{T_{j+1}, P_2, k} \circ P_2 \\ &= P_{2, T_{j+1}, k} \end{aligned}$$

4. In Σ'_j , the response from oracle $\mathcal{O}(x)$ is always equal to $E_k[P_1, P_2](x)$. When bad_2 occurs this is immediate. But if bad_2 does not occur then $x \neq s_1 \oplus k_1$ and $x \neq s_0 \oplus k_2$; we also know that $x \neq s_0 \oplus k_1 = x_{j+1}$ and $x \neq s_1 \oplus k_2 = x_{j+1}$ by assumption. So, $P_1^{(1)}(x \oplus k_1) = P_1(x \oplus k_1)$ and $P_2^{(1)}(x \oplus k_2) = P_2(x \oplus k_2)$.

Syntactically rewriting Σ'_j using the above observations yields an experiment that is identical to $\hat{\mathbf{H}}_j$. Thus using Eq. (10) we have

$$| \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \leq \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}}. \quad (12)$$

Hence, using Eq. (8), Eq. (9) and Eq. (12) we have

$$\begin{aligned} &| \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \\ &\leq | \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\Gamma_j) \rightarrow 1] | + | \Pr[\mathcal{A}(\Gamma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] | \\ &\quad + | \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \\ &\leq 4\sqrt{\frac{p}{2^n}} + 4\sqrt{\frac{p}{2^n}} + \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}} \\ &= \frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}}. \end{aligned}$$

5 Post-Quantum Security of KSoP12

In this section, we formally state and prove the post-quantum security of the KSoP12 construction.

Theorem 2. For any adversary $\mathcal{A}$ that makes q classical queries to KSoP12, i.e., $E[P]$ and p quantum queries to P (and its inverse) in total, it holds that

$$\text{Adv}_{\text{KSoP12}}^{\text{pq-prf}}(\mathcal{A}) \leq \frac{2q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.$$

The proof follows closely the line of argument used in [Theorem 1](#). As before, we consider a sequence of hybrid experiments indexed by j , denoted $(\mathbf{H}_j)$, in which the first j classical queries are answered using random functions. To hop between two consecutive games $\mathbf{H}_j$ and $\mathbf{H}_{j+1}$, we use an intermediate hybrid $\hat{\mathbf{H}}_j$. The analysis proceeds in two steps: we first apply the reprogramming lemma ([Lemma 1](#)) and subsequently invoke our extended resampling lemma for $r = 2$ swaps ([Lemma 3](#)). We retain all notation from the proof of [Theorem 1](#).

First, we define the reprogrammed permutation corresponding to P . For any $i \in [j]$, we sequentially choose random values $u_i \in \{0, 1\}^n \setminus (\mathcal{U} \cup \mathcal{Y} \cup \mathcal{P}_1 \cup \mathcal{P}_2)$, where $\mathcal{U} = \{u_1, \dots, u_{i-1}\}$, $\mathcal{Y} = \{y_i \oplus u_1 \oplus y_1, \dots, y_i \oplus u_{i-1} \oplus y_{i-1}\}$, $\mathcal{P}_1 = \{P(x_1 \oplus k_1), \dots, P(x_i \oplus k_1)\}$, and $\mathcal{P}_2 = \{P(x_1 \oplus k_2) \oplus y_i, \dots, P(x_i \oplus k_2) \oplus y_i\}$. Now, the corresponding reprogramming operator is defined as

$$\vec{S}_{T_j, P, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P(x_i \oplus k_1), u_i} \circ \text{swap}_{P(x_i \oplus k_2), (y_i \oplus u_i)}.$$

We define the reprogrammed permutation $P_{T_j, k}$ by

$$P_{T_j, k} = \vec{S}_{T_j, P, k} \circ P.$$

Now, let $\mathbf{H}_0, \dots, \mathbf{H}_q$ denote a sequence of $q + 1$ hybrids, where $\mathbf{H}_j$ is defined as follows:

1. Sample $k = (k_1, k_2) \xleftarrow{\$} \{0, 1\}^{2n}$.
2. Answer the first j classical queries of $\mathcal{A}$ using F and the quantum queries of $\mathcal{A}$ in the first j stages (which ends when the $(j + 1)$ -th classical query is made) using P .
3. Answer all subsequent classical queries from $\mathcal{A}$ using $E_k[P]$ and all subsequent quantum queries from $\mathcal{A}$ using $P_{T_j, k}$.

Thus, we have

$$\text{Adv}_{\text{KSoP12}}^{\text{pq-prf}}(\mathcal{A}) = |\Pr[\mathcal{A}(\mathbf{H}_0) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_q) \rightarrow 1]|. \quad (13)$$

Now for any $j \in \{0, \dots, q\}$, $\mathbf{H}_j$ can be viewed as a string of oracle calls made by $\mathcal{A}$

$$\underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \underbrace{(E_k[P], P_{T_j, k})^{q-j}}_{q-j \text{ classical queries}}.$$

where each occurrence of F (res. $E_k[P]$) indicates a single classical query and each occurrence of P (res. $P_{T_j, k}$) indicates the sequence of quantum primitive queries. To bound the distinguishing advantage between $\mathbf{H}_j$ and $\mathbf{H}_{j+1}$, we introduce an intermediate hybrid $\hat{\mathbf{H}}_j$ which is defined as follows:

1. Sample $k = (k_1, k_2) \xleftarrow{\$} \{0, 1\}^{2n}$.
2. Answer the first $j + 1$ classical queries of $\mathcal{A}$ using F and the quantum queries of $\mathcal{A}$ in the first j stages (which ends when the $(j + 1)$ -th classical query is made) using P .
3. Answer all subsequent classical queries from $\mathcal{A}$ using $E_k[P]$ and all subsequent quantum queries from $\mathcal{A}$ using $P_{T_{j+1}, k}$.

Now, we compare the following hybrids in pairs from top to bottom:

$$\begin{aligned}
\mathbf{H}_j &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad E_k[P], P_{T_j, k}, \quad \underbrace{(E_k[P], P_{T_j, k})^{q-j-1}}_{q-j-1 \text{ classical queries}} \\
\hat{\mathbf{H}}_j &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad F, \quad P_{T_{j+1}, k}, \quad \underbrace{(E_k[P], P_{T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}} \\
\mathbf{H}_{j+1} &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad F, \quad P, \quad \underbrace{(E_k[P], P_{T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}
\end{aligned}$$

In Lemma 7 and Lemma 8, we establish the bounds on the distinguishability of $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, as well as $\hat{\mathbf{H}}_j$ and $\mathbf{H}_{j+1}$. Thus using these two lemmas and Eq. (13), we have

$$\begin{aligned}
\text{Adv}_{\text{KSoP12}}^{\text{pq-prf}}(\mathcal{A}) &= |\Pr[\mathcal{A}(\mathbf{H}_0) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_q) \rightarrow 1]| \\
&\leq \sum_{j=0}^{q-1} \left(|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \right. \\
&\quad \left. + |\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \right) \\
&\leq \sum_{j=0}^{q-1} \left(\frac{2q}{2^n} + 8\sqrt{2}\sqrt{\frac{p}{2^n}} + 2p_{j+1}\sqrt{\frac{2(j+1)}{2^n}} \right) \\
&\leq \frac{2q^2}{2^n} + 8q\sqrt{2}\sqrt{\frac{p}{2^n}} + 2p\sqrt{\frac{2q}{2^n}} \\
&\leq \frac{2q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.
\end{aligned}$$

To complete the rest of the proof, it remains to bound the distinguishing advantage between $\hat{\mathbf{H}}_j$ and $\mathbf{H}_{j+1}$, as well as between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$. Proof of the following two lemmas are deferred to Section A.1 and Section A.2.

Lemma 7. For $j = 0, \dots, q-1$, we have

$$|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \leq 2p_{j+1}\sqrt{\frac{2(j+1)}{2^n}},$$

where p_{j+1} is the expected number of quantum queries $\mathcal{A}$ makes in the $(j+1)$ -th stage of the game.

Lemma 8. For $j = 0, \dots, q-1$, we have

$$|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \leq \frac{2q}{2^n} + 8\sqrt{2}\sqrt{\frac{p}{2^n}},$$

where q and p are the expected number of classical and quantum queries made by $\mathcal{A}$.

6 Post-Quantum Security of KSoP21

In this section we state and prove the formal post-quantum security result for the KSoP21 construction.

Theorem 3. For any adversary $\mathcal{A}$ that makes q classical queries to KSoP21, i.e., $E[P_1, P_2]$ and p quantum queries to P_1, P_2 (and their inverses) in total, it holds that

$$\text{Adv}_{\text{KSoP21}}^{\text{pq-prf}}(\mathcal{A}) \leq \frac{q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.$$

The proof proceeds along the same lines as the proof of [Theorem 1](#). The proof of [Theorem 3](#) mirrors that of [Theorem 1](#), with the only difference being the key structure. In the previous one, we consider $(k_1, k_2) \in \{0, 1\}^{2n}$, whereas here both k_1 and k_2 are replaced by a single key $k \in \{0, 1\}^n$. We retain all notation from the proof of [Theorem 1](#), with the sole modification that each occurrence of k_1, k_2 is replaced by k .

Here, we consider the following sequence of hybrid experiments, which we compare pairwise in a top-to-bottom manner.

$$\begin{aligned}
\mathbf{H}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{E_k[P_1, P_2], P_{1,T_j,k}, P_{2,T_j,k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_j,k}, P_{2,T_j,k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\
\hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{F, P_{1,T_{j+1},k}, P_{2,T_{j+1},k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\
\mathbf{H}_{j+1} &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \underbrace{F, P_1, P_2}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1,T_{j+1},k}, P_{2,T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}.
\end{aligned}$$

In [Lemma 9](#) and [Lemma 10](#), we establish the bounds on the distinguishability of $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, as well as $\hat{\mathbf{H}}_j$ and $\mathbf{H}_{j+1}$. Thus, using these two lemmas, we have

$$\begin{aligned}
\text{Adv}_{\text{KSoP21}}^{\text{pq-prf}}(\mathcal{A}) &= |\Pr[\mathcal{A}(\mathbf{H}_0) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_q) \rightarrow 1]| \\
&\leq \sum_{j=0}^{q-1} \left(|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \right. \\
&\quad \left. + |\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \right) \\
&\leq \sum_{j=0}^{q-1} \left(\frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}} + 2p_{j+1}\sqrt{\frac{2(j+1)}{2^n}} \right) \\
&\leq \frac{q^2}{2^n} + 12q\sqrt{\frac{p}{2^n}} + 2p\sqrt{\frac{2q}{2^n}} \\
&\leq \frac{q^2}{2^n} + \frac{3p\sqrt{q} + 12q\sqrt{p}}{2^{n/2}}.
\end{aligned}$$

To complete the rest of the proof, it remains to bound the distinguishing advantage between $\hat{\mathbf{H}}_j$ and $\mathbf{H}_{j+1}$, as well as between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$. Proof of the following two lemmas are deferred to [Section A.3](#) and [Section A.4](#).

Lemma 9. *For $j = 0, \dots, q-1$, we have*

$$|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \leq 2p_{j+1}\sqrt{\frac{2(j+1)}{2^n}},$$

where p_{j+1} is the expected number of quantum queries $\mathcal{A}$ makes in the $(j+1)$ -th stage of the game.

Lemma 10. *For $j = 0, \dots, q-1$, we have*

$$|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \leq \frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}}.$$

where q and p are the expected number of classical and quantum queries made by $\mathcal{A}$.

7 Key Recovery Attack on KSoP and Its Siblings

In this section, we give key-recovery attacks on KSoP and its variants under the Q1 model. Here, we follow the attack strategy introduced in 2019 by Bonnetain et al. [7].

7.1 Key Recovery Attack on KSoP

Consider u such that $0 \leq u \leq n$. We express k_1 as $k_1^l || k_1^r$ where $k_1^l \in \{0, 1\}^u$ and $k_1^r \in \{0, 1\}^{n-u}$. Similarly, we express k_2 as $k_2^l || k_2^r$. Overall key $k = (k_1, k_2)$ is of size $2n$.

We consider two functions, F and G , which are constructed based on the KSoP encryption and permutations as follows:

$$F : \{0, 1\}^{n-u} \times \{0, 1\}^n \times \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad F(i_1, i_2, x) := P_1(x || i_1) \oplus P_2(x || 0^{n-u} \oplus i_2)$$

and

$$G : \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad G(x) := \text{KSoP}(x || 0^{n-u}).$$

Observe that $F(i_1, i_2, x) \oplus G(x)$ has period k_1^l when $i_1 = k_1^r$ and $i_2 = (k_1^l \oplus k_2^l) || k_2^r$:

$$\begin{aligned} & F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x) \oplus G(x) \\ &= P_1(x || k_1^r) \oplus P_2(x || 0^{n-u} \oplus (k_1^l \oplus k_2^l) || k_2^r) \oplus \text{KSoP}(x || 0^{n-u}) \\ &= P_1(x || k_1^r) \oplus P_2(x \oplus k_1^l \oplus k_2^l || k_2^r) \oplus P_1(x || 0^{n-u} \oplus k_1) \oplus P_2(x || 0^{n-u} \oplus k_2) \\ &= P_1(x || k_1^r) \oplus P_2(x \oplus k_1^l \oplus k_2^l || k_2^r) \oplus P_1(x \oplus k_1^l || k_2^r) \oplus P_2(x \oplus k_2^l || k_2^r) \\ &= P_1(x \oplus k_1^l || k_2^r) \oplus P_2(x \oplus k_2^l || k_2^r) \oplus P_1(x || k_1^r) \oplus P_2(x \oplus k_1^l \oplus k_2^l || k_2^r) \\ &= F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x \oplus k_1^l) \oplus G(x \oplus k_1^l). \end{aligned}$$

Now we can recover a part of the key $k = (k_1, k_2)$ using Grover's search and rest of it using Simon's algorithm in Q1 with a high probability by performing $D = \mathcal{O}(2^u)$ classical queries to KSoP and making $T = \mathcal{O}(n^3 \cdot 2^{(2n-u)/2})$ time. The tradeoff is $DT^2 = \tilde{\mathcal{O}}(2^{2n})$, which balances at $T = D = \tilde{\mathcal{O}}(2^{2n/3})$. Besides, our attack requires $\mathcal{O}(n)$ classical memory and $\text{poly}(n)$ many qubits.

The detailed description of the key-recovery attack is identical to the one used in [7]. For completeness we provide the details in Appendix B.

Remark 1. In [15], Sun et al. present key-recovery attacks on permutation-based PRFs, as classified by Chen et al. [8]. In particular, they provide a generic key-recovery attack against two-permutation-based PRFs with complexity $\mathcal{O}(2^{2n/3})$. Although, KSoP belongs to this category, we independently develop the above dedicated key-recovery attack tailored specifically to the KSoP construction.

7.2 Key Recovery Attack on KSoP21

In this subsection, we discuss key recovery attack on KSoP21 under the Q1 model. Again, we follow the strategy of Bonnetain et al. Consider u such that $0 \leq u \leq n$. We express k as $k^l || k^r$ where $k^l \in \{0, 1\}^u$ and $k^r \in \{0, 1\}^{n-u}$. Overall key $k = (k^l || k^r)$ is of size n .

We consider two functions, F and G , which are constructed based on the KSoP21 encryption and permutations as follows:

$$F : \{0, 1\}^{n-u} \times \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad F(i, x) := P_1(x || i) \oplus P_2(x || i)$$

and

$$G : \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad G(x) := \text{KSoP21}(x || 0^{n-u})$$

Observe that $F(i, x) \oplus G(x)$ has period k^l when $i = k^r$:

$$\begin{aligned} & F(k^r, x) \oplus G(x) \\ &= P_1(x || k^r) \oplus P_2(x || k^r) \oplus \text{KSoP21}(x || 0^{n-u}) \\ &= P_1(x || k^r) \oplus P_2(x || k^r) \oplus P_1((x \oplus k^l) || k^r) \oplus P_2((x \oplus k^l) || k^r) \\ &= P_1((x \oplus k^l) || k^r) \oplus P_2((x \oplus k^l) || k^r) \oplus P_1(x || k^r) \oplus P_2(x || k^r) \\ &= F(k^r, x \oplus k^l) \oplus G(x \oplus k^l). \end{aligned}$$

Now we can recover a part of the key $k = (k_1, k_2)$ using Grover's search and rest of it using Simon's algorithm in Q1 with a high probability by performing $D = \mathcal{O}(2^u)$ classical queries to KSoP21 and making $T = \mathcal{O}(n^3 2^{(n-u)/2})$ time. The tradeoff is $DT^2 = \tilde{\mathcal{O}}(2^n)$, which balances at $T = D = \tilde{\mathcal{O}}(2^{n/3})$. Besides, our attack requires $\mathcal{O}(n)$ classical memory and $\text{poly}(n)$ many qubits.

7.3 Key Recovery Attack on KSoP12

In this section, we demonstrate full key recovery attack on KSoP12 under the Q1 model. Consider u such that $0 \leq u \leq n$. We express k_1 express as $k_1^l || k_1^r$ where $k_1^l \in \{0, 1\}^u$ and $k_1^r \in \{0, 1\}^{n-u}$. Similarly we express k_2 as $k_2^l || k_2^r$. Overall key $k = (k_1, k_2)$ is of size $2n$.

We consider two functions, F and G , which are constructed based on the KSoP12 encryption and permutations as follows:

$$F : \{0, 1\}^{n-u} \times \{0, 1\}^n \times \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad F(i_1, i_2, x) := P(x || i_1) \oplus P(x || 0^{n-u} \oplus i_2)$$

and

$$G : \{0, 1\}^u \rightarrow \{0, 1\}^n, \quad G(x) := \text{KSoP12}(x || 0^{n-u}).$$

Observe that $F(i_1, i_2, x) \oplus G(x)$ has period k_1^l when $i_1 = k_1^r$ and $i_2 = (k_1^l \oplus k_2^l) || k_2^r$:

$$\begin{aligned} & F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x) \oplus G(x) \\ &= P(x || k_1^r) \oplus P(x || 0^{n-u} \oplus (k_1^l \oplus k_2^l) || k_2^r) \oplus \text{KSoP12}(x || 0^{n-u}) \\ &= P(x || k_1^r) \oplus P(x \oplus k_1^l \oplus k_2^l || k_2^r) \oplus P(x || 0^{n-u} \oplus k_1) \oplus P(x || 0^{n-u} \oplus k_2) \\ &= P(x || k_1^r) \oplus P(x \oplus k_1^l \oplus k_2^l || k_2^r) \oplus P(x \oplus k_1^l || k_2^r) \oplus P(x \oplus k_2^l || k_2^r) \\ &= P(x \oplus k_1^l || k_2^r) \oplus P(x \oplus k_2^l || k_2^r) \oplus P(x || k_1^r) \oplus P(x \oplus k_1^l \oplus k_2^l || k_2^r) \\ &= F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x \oplus k_1^l) \oplus G(x \oplus k_1^l). \end{aligned}$$

Now we can recover a part of the key $k = (k_1, k_2)$ using Grover's search and rest of it using Simon's algorithm in Q1 with a high probability by performing $D = \mathcal{O}(2^u)$ classical queries to KSoP12 and making $T = \mathcal{O}(n^3 2^{(2n-u)/2})$ time. The tradeoff is $DT^2 = \tilde{\mathcal{O}}(2^{2n})$, which balances at $T = D = \tilde{\mathcal{O}}(2^{2n/3})$. Besides, our attack requires $\mathcal{O}(n)$ classical memory and $\text{poly}(n)$ many qubits.

8 Conclusion

In this paper, we studied the post-quantum security of the keyed sum-of-permutation construction and its related variants, including the single-permutation and single-key versions. We proved that all these constructions achieve $n/3$ -bit security in the Q1 model. In addition, we presented a key-recovery attack on KSoP21 with data and time complexity $\mathcal{O}(2^{n/3})$, thereby establishing its tight post-quantum security. For KSoP and KSoP12, we showed that key-recovery attacks can be carried out with data and time complexity $\mathcal{O}(2^{2n/3})$. Determining tight security bounds for these two constructions remains an open and interesting problem. Another promising direction is the analysis of the post-quantum security of the domain-separated single-key, single-permutation variant of KSoP. Also, extending the Q1 security analysis to the generalized keyed sum-of-permutations construction, where the sum involves r keyed permutations for arbitrary r , is an important open problem for future research.

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Supplementary Materials

A Deferred Proofs

A.1 Proof of Lemma 7

Recall we can have the following string of oracle call representations for the two hybrids.

$$\begin{array}{llll}
 \hat{\mathbf{H}}_j : & \underbrace{\mathbf{P}, (\mathbf{F}, \mathbf{P})^j}_{j \text{ classical queries}}, & \mathbf{F}, & \underbrace{\mathbf{P}_{T_{j+1},k}, (\mathbf{E}_k[\mathbf{P}], \mathbf{P}_{T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\
 \mathbf{H}_{j+1} : & \underbrace{\mathbf{P}, (\mathbf{F}, \mathbf{P})^j}_{j \text{ classical queries}}, & \mathbf{F}, & \underbrace{\mathbf{P}, (\mathbf{E}_k[\mathbf{P}], \mathbf{P}_{T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}.
 \end{array}$$

It is clear that the two hybrids are identical, except for the $(j+1)$ -th stages, where $\hat{\mathbf{H}}_j$ uses reprogrammed versions of the permutation $\mathbf{P}$, and $\mathbf{H}_{j+1}$ uses the original permutation $\mathbf{P}$. We will show that the distinguishing advantage of these two hybrids can be captured by the *reprogramming* bounds of the permutations.

More formally, given the adversary $\mathcal{A}$ we construct an ARP distinguisher $\mathcal{D}$ as follows (note that Phases 1, 2, 3 correspond to Phases 1, 2, 3 in the ARP game respectively).

Phase 1. For all the queries up to and including the $(j+1)$ -th classical query:

1. $\mathcal{D}$ samples uniform random permutations $\mathbf{P} \leftarrow \mathcal{P}(n)$ and uniform random function $\mathbf{F} \leftarrow \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using $\mathbf{F}$. Let, the resulting transcript of classical queries and responses is denoted by $T_{j+1} = \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$.
3. All quantum queries from $\mathcal{A}$ are answered using $\mathbf{P}$. More formally, $\mathcal{D}$ defines $F(t, x) := \mathbf{P}^t(x)$, for $t \in \{1, -1\}$.
4. $\mathcal{D}$ defines the randomized algorithm $\mathcal{B}$, which takes a key k , computes a set B of input/output pairs to be reprogrammed so that $F^{(B)}(t, x) = \mathbf{P}_{T_{j+1},k}^t(x)$, for all t, x .

Phase 2. For all the quantum queries in the $(j+1)$ -th stage:

1. $\mathcal{B}$ is run to generate the set B and $\mathcal{D}$ is given quantum access to F_b , where $F_0 = F(= \mathbf{P}^t)$ and $F_1 = F^{(B)}(= \mathbf{P}_{T_{j+1},k}^t)$.
2. All quantum queries from $\mathcal{A}$ are answered using F_b .

Phase 3. For all the queries beyond and including $(j+2)$ -th classical query:

1. $\mathcal{D}$ is given the randomness (specifically the key k here) used by $\mathcal{B}$ to generate B .
2. All classical queries from $\mathcal{A}$ are answered using $\mathbf{E}_k[\mathbf{P}]$.
3. All quantum queries from $\mathcal{A}$ are answered using F_1 .
4. Finally, $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARP distinguisher. If $b = 0$ (i.e., $\mathcal{D}$'s oracle in **Phase 2** is $F_0 = F$), then the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_{j+1}$. On the other hand, if $b = 1$ (i.e., $\mathcal{D}$'s oracle in **Phase 2** is $F_1 = F^{(B)}$), then the output of $\mathcal{A}$ is identically distributed to its output in $\hat{\mathbf{H}}_j$. It follows that $|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]|$ is equal to the distinguishing advantage of $\mathcal{D}$ in the reprogramming game. To bound this quantity using Lemma 1, we bound the reprogramming probability ϵ and the expected number of queries made by $\mathcal{D}$ in **Phase 2** (when $F_0 = F$).

The reprogramming probability ϵ can be bounded using the definition of $P_{T_{j+1},k}$ and the fact that $F^{(B)}(t, x) = P_{T_{j+1},k}^t(x)$. Fixing P and T_{j+1} , the probability that any given (t, x) is reprogrammed is at most the probability (over k) that it is in the set:

$$\{(1, (x_i \oplus k_1)), (-1, P^{-1}(u_i)), (1, (x_i \oplus k_2)), (-1, P^{-1}(y_i \oplus u_i))\}_{i=1}^{j+1}$$

Here, for any (t, x) , either it is a forward query or a backward query to F . In both cases, probability that (t, x) is reprogrammed is upper bounded by $2(j+1)/2^n$. Hence, we get $\epsilon \leq 2(j+1)/2^n$.

The expected number of queries made by $\mathcal{D}$ in **Phase 2** when $F_0 = F$ is equal to the expected number of queries made by $\mathcal{A}$ in its $(j+1)$ -th stage, which is p_{j+1} . Thus, by [Lemma 1](#), we have

$$|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \leq 2p_{j+1} \sqrt{\frac{2(j+1)}{2^n}}.$$

A.2 Proof of [Lemma 8](#)

For bounding the distinguishing advantage between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, we will introduce an intermediate hybrid called Σ_j . Now, we bound the distinguishing advantages between the following hybrids in pairs from top to bottom.

$$\begin{aligned} \mathbf{H}_j &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad E_k[P], \quad P_{T_j,k}, \quad \underbrace{(E_k[P], P_{T_j,k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \Sigma_j &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad E_k[P^{(1)}], \quad P_{T_j,k}^{(1)}, \quad \underbrace{(E_k[P^{(1)}], P_{T_j,k}^{(1)})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \hat{\mathbf{H}}_j &: \underbrace{P, (F, P)^j}_{j \text{ classical queries}}, \quad F, \quad P_{T_{j+1},k}, \quad \underbrace{(E_k[P], P_{T_{j+1},k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

Let, $\mathcal{A}$ be an adversary of $\mathbf{H}_j$ and Σ_j . From $\mathcal{A}$, we can construct an ARS distinguisher $\mathcal{D}$ as follows.

Phase 1. For all the queries up to (immediately before) the $(j+1)$ -th classical query:

1. $\mathcal{D}$ is given quantum oracle access to permutation P and $\mathcal{D}$ samples a uniform random function $F \in \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P .

Phase 2. For all the queries beyond and including $(j+1)$ -th classical query:

1. $\mathcal{D}$ receives $s_0, s_1, s_2, s_3 \in \{0, 1\}^n$ and $\mathcal{D}$ has now quantum oracle access to $P^{(b)}$, where $P^{(0)} = P$ and $P^{(1)} = \text{swap}_{P(s_0), P(s_1)} \circ \text{swap}_{P(s_2), P(s_3)} \circ P$.
2. Let $\mathcal{D}$ receives the $(j+1)$ -th classical query from $\mathcal{A}$ which is x_{j+1} .
3. $\mathcal{D}$ sets $k_1 = x_{j+1} \oplus s_0$ and $k_2 = x_{j+1} \oplus s_2$, and $k = (k_1, k_2)$.
4. All classical queries from $\mathcal{A}$ are answered using $E_k[P^{(b)}]$.
5. All quantum queries from $\mathcal{A}$ are answered using $P_{T_j,k}^{(b)}$.
6. Finally $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARS distinguisher. If $b = 0$, then $\mathbf{P}^{(0)} = \mathbf{P}$, thus the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_j$. On the other hand, if $b = 1$, then the output of $\mathcal{A}$ is identically distributed to its output in Σ_j . Hence, the distinguishing advantage between the hybrids $\mathbf{H}_j$ and Σ_j is equal to the distinguishing advantage of $\mathcal{D}$ in the resampling game. Using our extended resampling lemma, i.e., [Lemma 3](#) for $r = 2$, we have

$$|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1]| \leq 4\sqrt{2}\sqrt{\frac{p}{2^n}}. \quad (14)$$

Now, consider the following two events:

1. bad_1 : $(s_1 \oplus k_1) \in \{x_1, \dots, x_j\}$ or $(s_3 \oplus k_2) \in \{x_1, \dots, x_j\}$.
2. bad_2 : In phase 2, $\mathcal{A}$ queries its classical oracle on $s_1 \oplus k_1$ or on $s_3 \oplus k_2$ i.e $s_1 \oplus k_1 \in \{x_{j+2}, \dots, x_q\}$ or $s_3 \oplus k_2 \in \{x_{j+2}, \dots, x_q\}$.

Since, $s_1 \oplus k_1 = s_1 \oplus s_0 \oplus x_{j+1}$ is uniform and $s_3 \oplus k_2 = s_3 \oplus s_2 \oplus x_{j+1}$ is also uniform, $\Pr[\text{bad}_1] \leq 2j/2^n$. Now for the event bad_2 consider the distinguisher $\mathcal{D}'$ that behaves identically to $\mathcal{D}$ in **Phase 1** and **Phase 2**, but then when $\mathcal{A}$ terminates outputs 1 iff event bad_2 occurred. When $b = 0$, the probability that bad_2 occurs is at most $2(q-j)/2^n$. Since, $\mathcal{D}'$ is a distinguisher for the resampling game, so, we get $\Pr[\text{bad}_2] \leq 2(q-j)/2^n + 4\sqrt{2}\sqrt{p/2^n}$.

Algorithm 2 Hybrid experiment Σ_j . Including the boxed statements results in Σ'_j .

- 1: $\mathbf{P} \xleftarrow{\$} \mathcal{P}(n)$, $\mathbf{F} \xleftarrow{\$} \mathcal{F}(n, n)$
 - 2: Run $\mathcal{A}$ with quantum access to $\mathbf{P}$ and classical access to $\mathbf{F}$, until $\mathcal{A}$ makes its $(j+1)$ -th classical query x_{j+1}
 - 3: $T_j := \{(x_1, y_1), \dots, (x_j, y_j)\}$
 - 4: $s_0, s_1, s_2, s_3 \xleftarrow{\$} \{0, 1\}^n$, $\mathbf{P}^{(1)} := \mathbf{P} \circ \text{swap}_{s_0, s_1} \circ \text{swap}_{s_2, s_3}$.
 - 5: $k_1 := s_0 \oplus x_{j+1}$, $k_2 := s_1 \oplus x_{j+1}$, $k := (k_1, k_2)$
 - 6: $y_{j+1} := \mathbf{E}_k[\mathbf{P}^{(1)}](x_{j+1})$
 - 7: $Q := \mathbf{P}_{T_j, k}^{(1)}$.
 - 8: Give y_{j+1} to $\mathcal{A}$ as the answer to its $(j+1)$ -th classical query
 - 9: $T_{j+1} := \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$
 - 10: $u_{j+1} := \mathbf{P}(s_1)$
 - 11: **if** $(s_1 \oplus k_1) \in \{x_1, \dots, x_j\} \vee (s_3 \oplus k_2) \in \{x_1, \dots, x_j\}$ **then**
 - 12: $\text{bad}_1 := \text{true}$
 - 13: $Q := \mathbf{P}_{T_{j+1}, k}$
 - 14: Continue running $\mathcal{A}$ with quantum access to Q
(The remaining classical queries are answered as follows:)
 - 15: $y := \mathbf{E}_k[\mathbf{P}^{(1)}](x)$
 - 16: **if** $x = (s_1 \oplus k_1) \vee x = (s_3 \oplus k_2)$ **then**
 - 17: $\text{bad}_2 := \text{true}$
 - 18: $y := \mathbf{E}_k[\mathbf{P}]$
 - 19: return y .
-

In [Algorithm 2](#), we described Σ_j and Σ'_j . Observe that, Σ_j and Σ'_j are identical until either bad_1 or bad_2 occur, and so we have

$$|\Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma'_j) \rightarrow 1]| \leq \Pr[\text{bad}_1 \vee \text{bad}_2] \leq \frac{2q}{2^n} + 4\sqrt{2}\sqrt{\frac{p}{2^n}}. \quad (15)$$

Now, we are arguing that Σ'_j is identical to $\hat{\mathbf{H}}_j$:

1. Up to (just before) the $(j+1)$ -th classical query Σ'_j is exactly identical to $\hat{\mathbf{H}}_j$.
2. In Σ'_j , the response of the $(j+1)$ -th classical query, y_{j+1} is defined as

$$\begin{aligned} y_{j+1} &= E_k[P^{(1)}](x_{j+1}) = P^{(1)}(x_{j+1} \oplus k_1) \oplus P^{(1)}(x_{j+1} \oplus k_2) \\ &= P^{(1)}(s_0) \oplus P^{(1)}(s_2) = P(s_1) \oplus P(s_3). \end{aligned} \quad (16)$$

So, the value y_{j+1} is uniformly distributed.

3. In Σ'_j , the oracle Q is always equal to $P_{T_{j+1},k}$. When bad_1 occurs this is immediate (since then Q is set to $P_{T_{j+1},k}$ in line 13). But if bad_1 does not occur then for $i = 1, \dots, j$ we have $x_i \oplus k_1 \neq s_0$ and $x_i \oplus k_1 \neq s_1$ (where the former is because $x_{j+1} \oplus k_1 = s_0$ but $x_i \neq x_{j+1}$ by assumption, and the latter is by definition of bad_1). Similarly, for $i = 1, \dots, j$ we have $x_i \oplus k_2 \neq s_3$ and $x_i \oplus k_2 \neq s_2$. So, $P^{(1)}(x_i \oplus k_1) = P(x_i \oplus k_1)$ and $P^{(1)}(x_i \oplus k_2) = P(x_i \oplus k_2)$, for $i = 1, \dots, j$. Thus

$$\begin{aligned} \vec{S}_{T_j, P^{(1)}, k} &\stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P^{(1)}(x_i \oplus k_1), u_i} \circ \text{swap}_{P^{(1)}(x_i \oplus k_2), (u_i \oplus y_i)} \\ &= \prod_{i=1}^j \text{swap}_{P(x_i \oplus k_1), u_i} \circ \text{swap}_{P(x_i \oplus k_2), (u_i \oplus y_i)} \\ &= \vec{S}_{T_j, P, k} \end{aligned}$$

Therefore,

$$\begin{aligned} Q &= P_{T_j, k}^{(1)} = \vec{S}_{T_j, P^{(1)}, k} \circ P^{(1)} \\ &= \vec{S}_{T_j, P, k} \circ \text{swap}_{P(s_0), P(s_1)} \circ \text{swap}_{P(s_2), P(s_3)} \circ P \\ &= \vec{S}_{T_j, P, k} \circ \text{swap}_{P(x_{j+1} \oplus k_1), P(s_1)} \circ \text{swap}_{P(x_{j+1} \oplus k_2), P(s_3)} \circ P \\ &= \vec{S}_{T_j, P, k} \circ \text{swap}_{P(x_{j+1} \oplus k_1), u_{j+1}} \circ \text{swap}_{P(x_{j+1} \oplus k_2), (u_{j+1} \oplus y_{j+1})} \circ P \\ &= \vec{S}_{T_{j+1}, P, k} \circ P \\ &= P_{T_{j+1}, k} \end{aligned}$$

4. In Σ'_j , the response from oracle $\mathcal{O}(x)$ is always equal to $E_k[P](x)$. When bad_2 occurs this is immediate. But if bad_2 does not occur then $x \neq s_1 \oplus k_1$ and $x \neq s_0 \oplus k_2$; we also know that $x \neq s_3 \oplus k_2 = x_{j+1}$ and $x \neq s_2 \oplus k_2 = x_{j+1}$ by assumption. So, $P^{(1)}(x \oplus k_1) = P(x \oplus k_1)$ and $P^{(1)}(x \oplus k_2) = P(x \oplus k_2)$.

Syntactically rewriting Σ'_j using the above observations yields an experiment that is identical to $\hat{\mathbf{H}}_j$. Thus using Eq. (10) we have

$$| \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \leq \frac{2q}{2^n} + 4\sqrt{2}\sqrt{\frac{p}{2^n}}. \quad (17)$$

Hence, using Eq. (14) and Eq. (17) we have:

$$\begin{aligned} &| \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \\ &\leq | \Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] | + | \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] | \\ &\leq 4\sqrt{2}\sqrt{\frac{p}{2^n}} + \frac{2q}{2^n} + 4\sqrt{2}\sqrt{\frac{p}{2^n}}. \\ &= \frac{2q}{2^n} + 8\sqrt{2}\sqrt{\frac{p}{2^n}}. \end{aligned}$$

A.3 Proof of Lemma 9

Recall that we can have the following string of oracle call representations for the two hybrids.

$$\begin{aligned} \hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \quad \mathbf{F}, \quad \mathbf{P}_{1, T_{j+1}, k}, \quad \mathbf{P}_{2, T_{j+1}, k}, \quad \underbrace{(E_k[P_1, P_2], P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{H}_{j+1} &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, \quad \mathbf{F}, \quad \mathbf{P}_1, \quad \mathbf{P}_2, \quad \underbrace{(E_k[P_1, P_2], P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

It is clear that the two hybrids are identical, except for the $(j+1)$ -th stages, where $\hat{\mathbf{H}}_j$ uses reprogrammed versions of the permutations P_1 and P_2 , and $\mathbf{H}_{j+1}$ uses the original permutations P_1 and P_2 . We will show that the distinguishing advantage of these two hybrids can be captured by the *reprogramming* bounds of the permutations.

More formally, given the adversary $\mathcal{A}$ we construct an ARP distinguisher $\mathcal{D}$ as follows (note that Phases 1, 2, 3 correspond to Phases 1, 2, 3 in the ARP game respectively).

Phase 1. For all the queries up to and including the $(j+1)$ -th classical query:

1. $\mathcal{D}$ samples uniform random permutations $P_1, P_2 \leftarrow \mathcal{P}(n)$ and uniform random function $F \leftarrow \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_{j+1} = \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 . More formally, $\mathcal{D}$ defines $F(\alpha, t, x) := P_\alpha^t(x)$, for $t \in \{1, -1\}$ and $\alpha \in \{1, 2\}$.
4. $\mathcal{D}$ defines the randomized algorithm $\mathcal{B}$, which takes a key k , computes a set B of input/output pairs to be reprogrammed so that $F^{(B)}(\alpha, t, x) = P_{\alpha, T_{j+1}, k}^t(x)$, for all α, t, x .

Phase 2. For all the quantum queries in the $(j+1)$ -th stage:

1. $\mathcal{B}$ is run to generate the set B and $\mathcal{D}$ is given quantum access to F_b , where $F_0 = F(= P_\alpha^t)$ and $F_1 = F^{(B)}(= P_{\alpha, T_{j+1}, k}^t)$.
2. All quantum queries from $\mathcal{A}$ are answered using F_b .

Phase 3. For all the queries beyond and including $(j+2)$ -th classical query:

1. $\mathcal{D}$ is given the randomness (specifically the key k here) used by $\mathcal{B}$ to generate B .
2. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1, P_2]$.
3. All quantum queries from $\mathcal{A}$ are answered using F_1 .
4. Finally, $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARP distinguisher. If $b = 0$ (i.e., $\mathcal{D}$'s oracle in Phase 2 is $F_0 = F$), then the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_{j+1}$. On the other hand, if $b = 1$ (i.e., $\mathcal{D}$'s oracle in Phase 2 is $F_1 = F^{(B)}$), then the output of $\mathcal{A}$ is identically distributed to its output in $\hat{\mathbf{H}}_j$. It follows that $|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]|$ is equal to the distinguishing advantage of $\mathcal{D}$ in the reprogramming game. To bound this quantity using Lemma 1, we bound the reprogramming probability ϵ and the expected number of queries made by $\mathcal{D}$ in Phase 2 (when $F_0 = F$).

The reprogramming probability ϵ can be bounded using the definition of $P_{\alpha, T_{j+1}, k}$ and the fact that $F^{(B)}(\alpha, t, x) = P_{\alpha, T_{j+1}, k}^t(x)$. Fixing P_1, P_2 and T_{j+1} , the probability that any given (α, t, x) is reprogrammed is at most the probability (over k) that it is in the set:

$$\{(1, 1, (x_i \oplus k)), (1, -1, P_1^{-1}(u_i)), (2, 1, (x_i \oplus k)), (2, -1, P_2^{-1}(y_i \oplus u_i))\}_{i=1}^{j+1}$$

Here, for any (α, t, x) , either it is a forward query or a backward query to F . In both cases, probability that (α, t, x) is reprogrammed is upper bounded by $2(j+1)/2^n$. Hence, we get $\epsilon \leq 2(j+1)/2^n$.

The expected number of queries made by $\mathcal{D}$ in Phase 2 when $F_0 = F$ is equal to the expected number of queries made by $\mathcal{A}$ in its $(j+1)$ -th stage, which is p_{j+1} . Thus, by Lemma 1, we have

$$|\Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{H}_{j+1}) \rightarrow 1]| \leq 2p_{j+1} \sqrt{\frac{2(j+1)}{2^n}}.$$

A.4 Proof of Lemma 10

For bounding the distinguishing advantage between $\mathbf{H}_j$ and $\hat{\mathbf{H}}_j$, we will introduce two intermediate hybrids called $\mathbf{\Gamma}_j, \mathbf{\Sigma}_j$. Now, we bound the distinguishing advantages between the following hybrids in pairs from top to bottom.

$$\begin{aligned} \mathbf{H}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, E_k[P_1, P_2], \underbrace{P_{1, T_j, k}, P_{2, T_j, k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1, T_j, k}, P_{2, T_j, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{\Gamma}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, E_k[P_1^{(1)}, P_2], \underbrace{P_{1, T_j, k}^{(1)}, P_{2, T_j, k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1^{(1)}, P_2], P_{1, T_j, k}^{(1)}, P_{2, T_j, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \mathbf{\Sigma}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, E_k[P_1^{(1)}, P_2^{(1)}], \underbrace{P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1^{(1)}, P_2^{(1)}], P_{1, T_j, k}^{(1)}, P_{2, T_j, k}^{(1)})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \\ \hat{\mathbf{H}}_j &: \underbrace{P_1, P_2, (F, P_1, P_2)^j}_{j \text{ classical queries}}, F, \underbrace{P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k}}_{q-j-1 \text{ classical queries}}, \underbrace{(E_k[P_1, P_2], P_{1, T_{j+1}, k}, P_{2, T_{j+1}, k})^{q-j-1}}_{q-j-1 \text{ classical queries}}. \end{aligned}$$

Let, $\mathcal{A}$ be an adversary of $\mathbf{H}_j$ and $\mathbf{\Gamma}_j$. From $\mathcal{A}$, we can construct an ARS distinguisher $\mathcal{D}$ as follows.

Phase 1. For all the queries up to (immediately before) the $(j+1)$ -th classical query:

1. $\mathcal{D}$ is given quantum oracle access to permutation P_1 and $\mathcal{D}$ samples an independent random permutation $P_2 \in \mathcal{P}(n)$ and uniform random function $F \in \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 .

Phase 2. For all the queries beyond and including $(j+1)$ -th classical query:

1. $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ and $\mathcal{D}$ has now quantum oracle access to $P_1^{(b)}$, where $P_1^{(0)} = P_1$ and $P_1^{(1)} = \text{swap}_{P_1(s_0), P_1(s_1)} \circ P_1$.
2. Let $\mathcal{D}$ receives the $(j+1)$ -th classical query from $\mathcal{A}$ which is x_{j+1} .
3. $\mathcal{D}$ sets $k = x_{j+1} \oplus s_0$.
4. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1^{(b)}, P_2]$.

5. All quantum queries from $\mathcal{A}$ are answered using $P_{1,T_j,k}^{(b)}$ and $P_{2,T_j,k}$ correspondingly.
6. Finally $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

It is clear that $\mathcal{D}$ is a valid ARS distinguisher. If $b = 0$, then $P_1^{(0)} = P_1$, thus the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{H}_j$. On the other hand, if $b = 1$, then the output of $\mathcal{A}$ is identically distributed to its output in $\mathbf{\Gamma}_j$. Hence, the distinguishing advantage between the hybrids $\mathbf{H}_j$ and $\mathbf{\Gamma}_j$ is equal to the distinguishing advantage of $\mathcal{D}$ in the resampling game. Using Lemma 2 we have

$$|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{\Gamma}_j) \rightarrow 1]| \leq 4\sqrt{\frac{p}{2^n}}. \quad (18)$$

Now, we bound the distinguishing advantage between the hybrids $\mathbf{\Gamma}_j$ and $\mathbf{\Sigma}_j$. Let $\mathcal{A}$ be an adversary between these two hybrids, we can construct an ARS distinguisher $\mathcal{D}$ as follows. **Phase 1.** For all the queries up to (immediately before) the $(j+1)$ -th classical query:

1. $\mathcal{D}$ is given quantum oracle access to permutation P_2 and $\mathcal{D}$ samples an independent random permutation $P_1 \in \mathcal{P}(n)$ and a random function $F \in \mathcal{F}(n, n)$.
2. All classical queries from $\mathcal{A}$ are answered using F . Let, the resulting transcript of classical queries and responses is denoted by $T_j = \{(x_1, y_1), \dots, (x_j, y_j)\}$.
3. All quantum queries from $\mathcal{A}$ are answered using P_1, P_2 .

Phase 2. For all the queries beyond and including $(j+1)$ -th classical query:

1. $\mathcal{D}$ receives $s_0, s_1 \in \{0, 1\}^n$ and $\mathcal{D}$ has now quantum oracle access to $P_2^{(b)}$, where $P_2^{(0)} = P_2$ and $P_2^{(1)} = \text{swap}_{P_2(s_0), P_2(s_1)} \circ P_2$.
2. Let $\mathcal{D}$ receives the $(j+1)$ -th classical query from $\mathcal{A}$ which is x_{j+1} .
3. $\mathcal{D}$ sets $k = x_{j+1} \oplus s_0$. $\mathcal{D}$ reprogram the permutation P_1 to get $P_1^{(1)}$.
4. All classical queries from $\mathcal{A}$ are answered using $E_k[P_1^{(1)}, P_2^{(b)}]$.
5. All quantum queries from $\mathcal{A}$ are answered using $P_{1,T_j,k}^{(1)}$ and $P_{2,T_j,k}^{(b)}$ correspondingly.
6. Finally $\mathcal{D}$ returns whatever $\mathcal{A}$ returns.

Clearly, $\mathcal{D}$ is a valid ARS distinguisher. If $b = 0$, then $P_2^{(0)} = P_2$, thus the output distribution of $\mathcal{A}$ is identical to its output distribution in $\mathbf{\Gamma}_j$. On the other hand, if $b = 1$, then the output of $\mathcal{A}$ is identically distributed to its output in $\mathbf{\Sigma}_j$. Hence, the distinguishing advantage between the hybrids $\mathbf{\Gamma}_j$ and $\mathbf{\Sigma}_j$ is equal to the distinguishing advantage of $\mathcal{D}$ in the resampling game. Using Lemma 2 we have

$$|\Pr[\mathcal{A}(\mathbf{\Gamma}_j) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{\Sigma}_j) \rightarrow 1]| \leq 4\sqrt{\frac{p}{2^n}}. \quad (19)$$

Now, consider the following two events:

1. **bad₁**: $(s_1 \oplus k) \in \{x_1, \dots, x_j\}$.
2. **bad₂**: In phase 2, $\mathcal{A}$ queries its classical oracle on $s_1 \oplus k$ i.e $s_1 \oplus k \in \{x_{j+2}, \dots, x_q\}$.

Algorithm 3 Hybrid experiment Σ_j . Including the boxed statements results in Σ'_j .

```

1:  $P_1, P_2 \xleftarrow{\$} \mathcal{P}(n), F \xleftarrow{\$} \mathcal{F}(n, n)$ 
2: Run  $\mathcal{A}$  with quantum access to  $P_1, P_2$  and classical access to  $F$ , until  $\mathcal{A}$  makes its
    $(j+1)$ -th classical query  $x_{j+1}$ 
3:  $T_j := \{(x_1, y_1), \dots, (x_j, y_j)\}$ 
4:  $s_0, s_1 \xleftarrow{\$} \{0, 1\}^n, P_1^{(1)} := P_1 \circ \text{swap}_{s_0, s_1}, P_2^{(1)} := P_2 \circ \text{swap}_{s_0, s_1}$ 
5:  $k := s_0 \oplus x_{j+1}$ .
6:  $y_{j+1} := E_k[P_1^{(1)}, P_2^{(1)}](x_{j+1})$ 
7:  $Q_1 := P_{1, T_j, k}^{(1)}, Q_2 := P_{2, T_j, k}^{(1)}$ 
8: Give  $y_{j+1}$  to  $\mathcal{A}$  as the answer to its  $(j+1)$ -th classical query
9:  $T_{j+1} := \{(x_1, y_1), \dots, (x_{j+1}, y_{j+1})\}$ 
10:  $u_{j+1} := P_1(s_1)$ 
11: if  $(s_1 \oplus k) \in \{x_1, \dots, x_j\}$  then
12:   bad1 := true
13:    $Q_1 := P_{1, T_{j+1}, k}$  and  $Q_2 := P_{2, T_{j+1}, k}$ 
14: Continue running  $\mathcal{A}$  with quantum access to  $Q_1, Q_2$ 
   (The remaining classical queries are answered as follows:)
15:  $y := E_k[P_1^{(1)}, P_2^{(1)}](x)$ 
16: if  $x = (s_1 \oplus k)$  then
17:   bad2 := true
18:    $y := E_k[P_1, P_2]$ 
19: return  $y$ .
```

Since, $s_1 \oplus k = s_1 \oplus s_0 \oplus x_{j+1}$ is uniform, $\Pr[\text{bad}_1] \leq j/2^n$. Now for the event bad_2 consider the distinguisher $\mathcal{D}'$ that behaves identically to $\mathcal{D}$ in **Phase 1** and **Phase 2**, but then when $\mathcal{A}$ terminates outputs 1 iff event bad_2 occurred. When $b = 0$, the probability that bad_2 occurs is at most $(q-j)/2^n$. Since, $\mathcal{D}'$ is a distinguisher for the resampling game, so, we get $\Pr[\text{bad}_2] \leq (q-j)/2^n + 4\sqrt{p/2^n}$.

In **Algorithm 3**, we described Σ_j and Σ'_j . Observe that, Σ_j and Σ'_j are identical until either bad_1 or bad_2 occur, and so we have

$$|\Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma'_j) \rightarrow 1]| \leq \Pr[\text{bad}_1 \vee \text{bad}_2] \leq \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}}. \quad (20)$$

Now, we are arguing that Σ'_j is identical to $\hat{\mathbf{H}}_j$:

1. Up to (just before) the $(j+1)$ -th classical query Σ'_j is exactly identical to $\hat{\mathbf{H}}_j$.
2. In Σ'_j , the response of the $(j+1)$ -th classical query, y_{j+1} is defined as

$$\begin{aligned} y_{j+1} &= E_k[P_1^{(1)}, P_2^{(1)}](x_{j+1}) = P_1^{(1)}(x_{j+1} \oplus k) \oplus P_2^{(1)}(x_{j+1} \oplus k) \\ &= P_1^{(1)}(s_0) \oplus P_2^{(1)}(s_0) = P_1(s_1) \oplus P_2(s_1). \end{aligned} \quad (21)$$

So, the value y_{j+1} is uniformly distributed.

3. In Σ'_j , the oracles Q_1 and Q_2 are always equal to $P_{1, T_{j+1}, k}$ and $P_{2, T_{j+1}, k}$ respectively. When bad_1 occurs this is immediate (since then Q_1 and Q_2 are set to $P_{1, T_{j+1}, k}$ and $P_{2, T_{j+1}, k}$ resp. in line 13). But if bad_1 does not occur then for $i = 1, \dots, j$ we have $x_i \oplus k \neq s_0$ and $x_i \oplus k \neq s_1$ (where the former is because $x_{j+1} \oplus k_1 = s_0$ but $x_i \neq x_{j+1}$

by assumption, and the latter is by definition of bad_1). So, $P_1^{(1)}(x_i \oplus k) = P_1(x_i \oplus k)$ and $P_2^{(1)}(x_i \oplus k) = P_2(x_i \oplus k)$, for $i = 1, \dots, j$. Thus

$$\vec{S}_{T_j, P_1^{(1)}, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_1^{(1)}(x_i \oplus k), u_i} = \prod_{i=1}^j \text{swap}_{P_1(x_i \oplus k), u_i} = \vec{S}_{T_j, P_1, k}$$

and

$$\vec{S}_{T_j, P_2^{(1)}, k} \stackrel{\text{def}}{=} \prod_{i=1}^j \text{swap}_{P_2^{(1)}(x_i \oplus k), (y_i \oplus u_i)} = \prod_{i=1}^j \text{swap}_{P_2(x_i \oplus k), (y_i \oplus u_i)} = \vec{S}_{T_j, P_2, k}$$

Therefore,

$$\begin{aligned} Q_1 &= P_{1, T_j, k}^{(1)} = \vec{S}_{T_j, P_1^{(1)}, k} \circ P_1^{(1)} = \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(s_0), P_1(s_1)} \circ P_1 \\ &= \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(x_{j+1} \oplus k), P_1(s_1)} \circ P_1 \\ &= \vec{S}_{T_j, P_1, k} \circ \text{swap}_{P_1(x_{j+1} \oplus k), u_{j+1}} \circ P_1 \\ &= \vec{S}_{T_{j+1}, P_1, k} \circ P_1 \\ &= P_{1, T_{j+1}, k} \end{aligned}$$

and

$$\begin{aligned} Q_2 &= P_{2, T_j, k}^{(1)} = \vec{S}_{T_j, P_2^{(1)}, k} \circ P_2^{(1)} = \vec{S}_{T_j, P_2, k} \circ \text{swap}_{P_2(s_0), P_2(s_1)} \circ P_2 \\ &= \vec{S}_{T_j, P_2, k} \circ \text{swap}_{P_2(x_{j+1} \oplus k), P_2(s_1)} \circ P_2 \\ &= \vec{S}_{T_j, P_2, k} \circ \text{swap}_{P_2(x_{j+1} \oplus k), (u_{j+1} \oplus y_{j+1})} \circ P_2 \\ &= \vec{S}_{T_{j+1}, P_2, k} \circ P_2 \\ &= P_{2, T_{j+1}, k} \end{aligned}$$

4. In Σ'_j , the response from oracle $\mathcal{O}(x)$ is always equal to $E_k[P_1, P_2](x)$. When bad_2 occurs this is immediate. But if bad_2 does not occur then $x \neq s_1 \oplus k$; we also know that $x \neq s_0 \oplus k = x_{j+1}$ by assumption. So, $P_1^{(1)}(x \oplus k) = P_1(x \oplus k)$ and $P_2^{(1)}(x \oplus k) = P_2(x \oplus k)$.

Syntactically rewriting Σ'_j using the above observations yields an experiment that is identical to $\hat{\mathbf{H}}_j$. Thus using Eq. (20) we have

$$|\Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \leq \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}}. \quad (22)$$

Hence, using Eq. (18), Eq. (19) and Eq. (22) we have

$$\begin{aligned} &|\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\hat{\mathbf{H}}_j) \rightarrow 1]| \\ &\leq |\Pr[\mathcal{A}(\mathbf{H}_j) \rightarrow 1] - \Pr[\mathcal{A}(\Gamma_j) \rightarrow 1]| + |\Pr[\mathcal{A}(\Gamma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma_j) \rightarrow 1]| \\ &\quad + |\Pr[\mathcal{A}(\Sigma_j) \rightarrow 1] - \Pr[\mathcal{A}(\Sigma'_j) \rightarrow 1]| \\ &\leq 4\sqrt{\frac{p}{2^n}} + 4\sqrt{\frac{p}{2^n}} + \frac{q}{2^n} + 4\sqrt{\frac{p}{2^n}} \\ &= \frac{q}{2^n} + 12\sqrt{\frac{p}{2^n}}. \end{aligned}$$

B Detailed Analysis of the KSoP Attack

We first prepare a quantum state $|\psi_G\rangle = \bigotimes_{j=1}^c \sum_{x_j \in \{0,1\}^u} |x_j\rangle |G(x_j)\rangle$. For that we introduce a quantum oracle U_p . For known value $G(p)$, that we get by making classical query to KSoP, for each $p \in \{0, 1, \dots, 2^{u-1}\}$, we define U_p as:

$$|p\rangle \bigotimes_{j=1}^c |x_j\rangle |y\rangle \rightarrow \begin{cases} |p\rangle \bigotimes_{j=1}^c |x_j\rangle |y \oplus G(x_j)\rangle, & \text{if } x_1 = x_2 = \dots = x_c = p, \\ |p\rangle \bigotimes_{j=1}^c |x_j\rangle |y\rangle, & \text{otherwise} \end{cases}$$

We use Algorithm 4 to generate $|\psi_G\rangle$:

Algorithm 4 For preparing $|\psi_G\rangle$

Input: classical query access to the KSoP, integer c .

Output: $|\psi_G\rangle$.

- 1: Initialize three registers $|0\rangle^{\otimes u} |0\rangle^{\otimes cu} |0\rangle^{\otimes cn}$.
- 2: Perform quantum gate $H^{\otimes cu}$ on the second register:

$$|\phi\rangle = \bigotimes_{j=1}^c \sum_{x_j \in \{0,1\}^u} |x_j\rangle |0\rangle.$$

- 3: **for** $p = 0, 1, \dots, 2^u - 1$ **do**
 - 4: Prepare the u -qubit state $|p\rangle$ on the first register.
 - 5: Compute $G(p)$ by making classical queries to the KSoP construction.
 - 6: Apply the quantum oracle U_p to $|p\rangle \otimes |\phi\rangle$.
 - 7: Uncompute $|p\rangle$.
 - 8: **return** $|\psi_G\rangle$.
-

We consider one more quantum oracle $O_F : (i_1, i_2, x) \mapsto |F(i_1, i_2, x)\rangle$. Now, we use Algorithm 5 to find $k_1^r || (k_1^l \oplus k_2^l) || k_2^r$.

Next, we consider $f : \{0, 1\}^u \rightarrow \{0, 1\}^n$, defined as $f(x) = F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x) \oplus G(x)$. We then apply Algorithm 6, i.e., Simon's algorithm for Q1, and under the condition that $\ell = \mathcal{O}(n)$, we obtain k_1^l , which is the period for f .

Complexity Analysis We analyze the complexity of the individual algorithms as follows.

Analysis of Algorithm 4: In Algorithm 4, $|\psi_G\rangle$ can be prepared using few gates only. Implementation of each U_p requires one classical query to $G(x)$. One such evaluation can be done in $\mathcal{O}(1)$ evaluation of KSoP. So, Algorithm 4 generates $|\psi_G\rangle$ with $\mathcal{O}(2^u)$ classical queries and $\mathcal{O}(u + c(n + u))$ qubits.

Analysis of Algorithm 5: This analysis has an online as well as an offline phase. Step 2 comprises of the online phase, where $|\psi_G\rangle$ is prepared. As we have seen in the of Bonnetain et al. [7], this $|\psi_G\rangle$ will be reused in each Grover iterations. Steps 3-9 comprises of the offline phase where specifically steps 4-8 corresponds to the *test* oracle:

$$|i\rangle |\psi_G\rangle |b\rangle \mapsto \begin{cases} |i\rangle |\psi_G\rangle |b \oplus 1\rangle, & \text{if } F(i, x) \oplus G(x) \text{ is periodic function,} \\ |i\rangle |\psi_G\rangle |b \oplus 0\rangle, & \text{otherwise} \end{cases}$$

where $i \in \{0, 1\}^{2n-u}$ and is also expressed as $i = i_1 || i_2$ with $i_1 \in \{0, 1\}^{n-u}$ and $i_2 \in \{0, 1\}^n$. Overall we make $2c$ quantum queries to oracle $F(i, x)$. Also, step 6 requires $\mathcal{O}(n^3)$ time fro computation. So, Algorithm 5 recovers $k_1^r || (k_1^l \oplus k_2^l) || k_2^r$ in $\mathcal{O}(2^u)$ classical queries and in $\mathcal{O}((n^3 + 2c)2^{\frac{(2n-u)}{2}})$ time.

Algorithm 5 Recover $k_1^r || (k_1^l \oplus k_2^l) || k_2^r$ using classical queries

Input: oracle $O_F : |i_1, i_2, x\rangle \mapsto |F(i_1, i_2, x)\rangle$ with $F : \{0, 1\}^{2n-u} \times \{0, 1\}^u \rightarrow \{0, 1\}^n$, classical query access to the KSoP construction, integer c .

Output: $k_1^r || (k_1^l \oplus k_2^l) || k_2^r$.

- 1: Initialize $|0\rangle^{\otimes(2n-u)} |0\rangle^{\otimes cu} |0\rangle^{\otimes cn} |1\rangle$.
- 2: Apply Algorithm 4 to attain

$$|\psi_G\rangle = \bigotimes_{j=1}^c \sum_{x_j \in \{0,1\}^u} |x_j\rangle |G(x_j)\rangle.$$

- 3: Perform $H^{\otimes(2n-u)} \otimes I_u^{\otimes c} \otimes I_n^{\otimes c} \otimes H$ to create

$$\sum_{i \in \{0,1\}^{2n-u}} |i\rangle \otimes |\psi_G\rangle |b\rangle,$$

where $|b\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and I_n denotes the identity on n qubits.

- 4: Make c oracle queries to O_F :

$$\sum_{i \in \{0,1\}^{2n-u}} |i\rangle \bigotimes_{j=1}^c \sum_{x_j \in \{0,1\}^u} |x_j\rangle |G(x_j) \oplus F(i, x_j)\rangle = \sum_{i \in \{0,1\}^{2n-u}} |i\rangle \otimes |\psi_{G \oplus F}\rangle |b\rangle.$$

- 5: Apply quantum gate $I_{2n-u} \otimes H^{\otimes cu} \otimes I_n^{\otimes c} \otimes I_1$:

$$\sum_{i \in \{0,1\}^{2n-u}} |i\rangle \bigotimes_{j=1}^c \sum_{x_j, y_j \in \{0,1\}^u} (-1)^{x_j \cdot y_j} |y_j\rangle |G(x_j) \oplus F(i, x_j)\rangle |b\rangle.$$

- 6: Compute the dimension d of the vector space spanned by $y_1, y_2, \dots, y_c$. If $d = u$, set $r = 0$; if $d < u$, set $r = 1$. Add r to b , then uncompute d and r to obtain

$$\sum_{i \in \{0,1\}^{2n-u}} |i\rangle \bigotimes_{j=1}^c \sum_{x_j, y_j \in \{0,1\}^u} (-1)^{x_j \cdot y_j} |y_j\rangle |G(x_j) \oplus F(i, x_j)\rangle |b \oplus r\rangle.$$

- 7: Reapply Hadamard gates $I_{2n-u} \otimes H^{\otimes cu} \otimes I_n^{\otimes c} \otimes I_1$.

- 8: Make c oracle queries to O_F :

$$\sum_{i \in \{0,1\}^{2n-u}} |i\rangle \otimes |\psi_G\rangle |b \oplus r\rangle.$$

- 9: After performing $O(2^{(2n-u)/2})$ Grover iterations, measure to return $k_1^r || (k_1^l \oplus k_2^l) || k_2^r$.
-

Algorithm 6 Simon's algorithm using classical queries

Input: Classical query access to the periodic function $f(x) : \{0, 1\}^u \rightarrow \{0, 1\}^n$ **Output:** s

- 1: Perform 2^u classical queries to $f(x)$ to prepare the quantum state

$$|\psi_f\rangle = \bigotimes_{j=1}^{\ell} \sum_{x \in \{0,1\}^u} |x\rangle |f(x)\rangle.$$

- 2: Perform Hadamard gates H^n on each state $|x\rangle$ to obtain

$$\bigotimes_{j=1}^{\ell} \sum_{x,v \in \{0,1\}^u} (-1)^{x \cdot v} |v\rangle |f(x)\rangle.$$

- 3: Measure all $|v\rangle$ registers to obtain ℓ vectors $v_1, v_2, \dots, v_\ell$.
- 4: Compute the dimension

$$d = \dim(\text{Span}(v_1, v_2, \dots, v_\ell)).$$

If $d \neq u - 1$, return $\perp$. If $d = u - 1$, compute a nonzero vector

$$s \in \{0, 1\}^u \quad \text{such that} \quad s \perp \text{Span}(v_1, v_2, \dots, v_\ell),$$

using classical linear algebra.

Analysis of Algorithm 6: Here we apply Algorithm 6 to $f(x) = F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x) \oplus G(x)$. We can recover k_1^l with $\mathcal{O}(2^u)$ classical queries, and in $\mathcal{O}(u^3 + u)$ time with $\mathcal{O}(u^2)$ qubits.

According to Theorem 2 of [6], $c = n + \tau + 1$ (τ is a positive integer) is enough to compute the unique $k_1^r, (k_1^l \oplus k_2^l) || k_2^r$. By performing Simon's algorithm in the Q1 model to the function $F(k_1^r, (k_1^l \oplus k_2^l) || k_2^r, x) \oplus G(x)$, we can recover k_1^l with $\mathcal{O}(2^u)$ classical queries, $\mathcal{O}(u^3)$ time and $\text{poly}(u)$ many qubits.